# Beyond Rhetoric: An Axiomatic Measure of Civilization for Politics and Diplomacy

### Subtitle: The Civic Reflexivity Index (CRI) and Protocols for Discussable Governance

## Abstract

This article proposes a procedure-first architecture for governing problems that are **self-referential**—where proposed policies change the very rules by which they are judged. We argue that many stalemates in climate governance, deterrence, financial stability, and AI safety are not failures of evidence but of **discussability**: claims lack shared observables, cross-scale maps, and agreed protocols once rules are endogenous. We therefore specify four axioms—**Reflexivity** (some agendas alter evaluation rules), **Observability** (claims must bind at least one shared **Collapse Observable**, CO), **Incompleteness** (some disputes require axiom expansion or domain rebasing), and **Delayed Collapse** (retain multiple candidates in non-phase-free regimes until alignment signals fire). Building on these, we define the **Civic Reflexivity Index (CRI)**—a six-component capacity measure: **sra** (explicit reflexivity), **csc** (cross-scale consistency), **co** (CO-kit coverage), **dcr** (use of delayed collapse), **gdn** (Gödel-Navigator loop handling), and **bhc** (black-hole coverage for currently non-discussables), each scored on anchored rubrics with published confidence. We introduce a "geometry of negotiation": minimal scale-maps with declared **invariants** and **losses**, curvature-based early-warning used responsibly (variance, recovery, AR(1) with persistence and triangulation), and three operational protocols—**Delayed-Collapse**, **Gödel-Navigator** (relax / detour / rebase), and a **Black-Hole Registry** that preserves accountability when cores must remain dark. Institutionally, we provide a public **CRI Dashboard**, treaty-grade procedure language, and an **Audit Pack** (checklists, logs, anti-gaming tests) that make decisions tamper-evident and reproducible. The framework is ideologically neutral: it does not pick ends; it binds any worldview to observables, gates, and audit trails that **travel across regimes and time**, turning debate into discussability and rhetoric into governance.

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# Part I — Why Civilization Needs Axioms

# The Rhetorical Trap

## Self-referential deadlocks: why "good reasons" cancel out

The most persistent public arguments are not short of reasons; they are short of **shared observables**. When parties argue policy that would rewrite the rules by which it is judged, each side can remain indefinitely right **within its own frame**. This is the **self-referential deadlock**: arguments loop because acceptance of a proposal would change the yardstick used to accept it.

Two patterns recur:

1. **Frame-locked symmetry.**
   Side A assesses a policy under today's weights and constraints; Side B assesses under tomorrow's—precisely the weights the policy seeks to install. Each can marshal impeccable evidence relative to its chosen frame; neither can defeat the other **across** frames.

2. **Invisible measurement.**
   Claims are advanced without agreed **Collapse Observables (COs)**—no shared events or thresholds that would count as narrowing the dispute. Without COs, reasons accumulate but never cash out.

In such conditions, "good reasons" cancel because they do not **intersect** in a measurement space. The debate is functional rhetoric, not governance.

## Policy as game-rule change: the hidden circularity

Policies are not mere moves on a fixed board; they are proposals to alter the board: payoffs, legal constraints, information flows, even who counts as a stakeholder. That makes policy **self-referential**. Evaluating a carbon price using market valuations, for example, ignores that the price will **rewrite** those valuations and the risk surface from which they came. Deterrence doctrines are justified by strategic stability metrics that the doctrines themselves will reshape. Bank backstops are judged by "market discipline" that backstops will partly disarm. AI guardrails are assessed by an oversight capacity the guardrails seek to build.

This circularity is not a flaw in politics; it is its nature. The flaw is pretending the circle is a line. When we treat rule-changing proposals as if they were moves inside unchanging rules, the debate becomes unfalsifiable performance.

### Timeless failures: climate, deterrence, financial contagion, AI

Consider four archetypes that recur across centuries and technologies:

- **Climate governance.**
  Discount rates, equity weights, and risk tails are not neutral; each is a **rule choice** with distributional and temporal consequences. Absent a declared CO kit (e.g., physical risk bands, adaptation loss functions, tail exceedance counts) and a cross-scale map (household ↔ firm ↔ state ↔ planet), "cost-benefit" exchanges become dueling frames.

- **Deterrence and arms stability.**
  Signaling works by changing the other side's beliefs about future rules of response. Without COs on alert postures, escalation gates, and verification windows, rhetoric about "resolve" substitutes for measurable stability.

- **Financial contagion.**
  Lender-of-last-resort and resolution regimes reorganize balance-sheet networks. If debate omits ex-ante triggers, loss-sharing observables, and liquidity stress COs, "moral hazard" vs "panic prevention" stalemates on principle.

- **AI alignment and safety.**
  Oversight, evaluation, and deployment standards are self-referential: they aim to create the governance capacity by which they will be judged. Without domain COs (capability ceilings, incident taxonomies, post-incident audits) and staged **Delayed Collapse** procedures, we oscillate between acceleration and paralysis.

In each case the surface dispute is about facts and values; the deep impasse is about **measurement under changing rules**.

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## Civilization as Reflexive Capacity

### From artifacts (GDP, tech) to governance reflexivity

A civilization is often credited for what it **builds**—roads, rockets, patents, GDP. Those are artifacts. The harder question is what a civilization can **unbuild and rebuild in its own rule space**. Do we possess routines to:

- name when a problem is **self-referential**,
- publish the **CO kit** that turns rhetoric into evidence,
- map micro ↔ meso ↔ macro effects with declared losses,
- declare "currently non-discussable" zones (black holes) without stigma, and
- in non-phase-free regimes, run **Delayed Collapse** until cross-scale consistency appears?

Call this capacity **governance reflexivity**. It is not a political ideology; it is a metacapability—a way to keep politics from confusing performance with decision.

### A civilization test: can we make hard problems discussable?

A timeless test fits on one page and works in any era:

1. **Declare reflexivity.**
   For each major agenda, state whether adopting it would change the metrics by which it is judged. If yes, you are in the self-referential class.

2. **Publish the CO kit.**
   List the minimal set of observables that would count as narrowing disagreement (events, thresholds, gates), and who collects them.

3. **Map scales.**
   Show the micro ↔ meso ↔ macro correspondences and note what is lost in translation. (If you do not disclose losses, you have not mapped scales.)

4. **Register black holes.**
   Name what is **currently non-discussable**—ethical axioms, sacred constraints, strategic secrets—and how risks from those zones are mitigated.

5. **Commit to Delayed Collapse.**
   If phase order matters, retain at least two live candidates and specify the alignment signals that will justify collapse to one.

A polity that can do these five things for its hardest problems is civilized in a way that outlives fashions, ideologies, and technologies. It has learned to **make hard problems discussable**—not by winning the argument, but by building the measurement.

That is why we need axioms. They are not dogmas. They are the minimal, reusable contracts that turn performance into governance, and governance into something a future historian can audit without guessing what we "really meant." In the parts that follow, we make these contracts explicit: the observables, the index, and the protocols that keep debate from looping and power from hiding in ambiguity.

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# Part II — The Axiomatic Core

# Postulates, Not Preferences

### A1–A4 stated and motivated

- **A1 (Reflexivity).**
  *Statement*. Some proposals change payoffs, constraints, or information rules that determine their own evaluation.
  *Motivation*. Without acknowledging this, parties talk past one another: each "wins" inside a frame the proposal itself would alter.

- **A2 (Observability).**
  *Statement*. A claim is discussable only if it binds at least one **Collapse Observable (CO)**—a measurable event/threshold whose occurrence narrows disagreement.
  *Motivation*. Reasons without shared measurement cannot converge; COs make rhetoric cash-out into evidence.

- **A3 (Incompleteness).**
  *Statement*. Some disputes are undecidable within current rules. They require **axiom expansion** or **domain rebasing** to proceed.
  *Motivation*. This prevents infinite loops masquerading as deliberation and legitimizes structured rule changes.

- **A4 (Delayed Collapse).**
  *Statement*. When the order/timing of moves changes outcomes (non-phase-free), retain ≥2 candidates and delay commitment until cross-scale alignment is evidenced.
  *Motivation*. Early lock-in creates path dependence and brittle policy; DC trades speed for stability.

### Why axioms, not ideologies: neutrality, repeatability, auditability

- **Neutrality.** Axioms define *when and how* to measure, not *what to believe*. Any ideology can play—provided it binds COs.
- **Repeatability.** Axioms make competing analyses comparable; COs and protocols can be rerun by others.
- **Auditability.** Decisions leave verifiable traces: CO hits/misses, loop logs, DC gates, and BHR entries.

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# Collapse Observables (CO): Making Claims Testable

## Definition and minimal kits by domain

A **CO** is a pre-declared, auditable observation that—if realized—reduces the live policy space. A minimal **CO kit** lists 5–9 such observables plus collection procedures.

**Design requirements (all must hold):**

1. **Pre-registration.** Defined *before* analysis or negotiation begins.
2. **Third-party auditable.** Data sources and methods traceable; time-stamped.
3. **Frame-bridging.** Each CO narrows the gap between at least two rival frames.
4. **Non-perverse.** Monitored for gaming; include anti-manipulation checks.
5. **Cross-scale hooks.** Each CO maps to micro, meso, and/or macro effects (declaring losses).

**Minimal kits (illustrative, not exhaustive):**

- **Climate governance**
  $CO_1$ tail exceedance counts (e.g., >X ppm-years above threshold)
  $CO_2$ adaptation loss index (infrastructure/service-level failures)
  $CO_3$ sectoral abatement realized vs. plan (audited tons)
  $CO_4$ risk-weighted discount update trigger (when damage tails widen)
  $CO_5$ equity-weight disclosure (which groups bear residual risk)
- **Deterrence/arms stability**
  $CO_1$ alert-posture transitions (with verification windows)
  $CO_2$ exercise-to-notification ratios (signaling clarity)
  $CO_3$ escalation gate adherence/violations
  $CO_4$ near-miss incident taxonomy counts
  $CO_5$ hot-line latency distributions during crises
- **Financial contagion**
  $CO_1$ liquidity stress breaches (pre-declared metrics)
  $CO_2$ resolution waterfall execution (loss-sharing order)
  $CO_3$ circuit-breaker triggers & resets logged
  $CO_4$ interbank spread dynamics post-intervention
  $CO_5$ backstop sunset compliance
- **AI governance**
  $CO_1$ capability ceiling tests (pre-committed evals)
  $CO_2$ deployment-stage gates passed/failed
  $CO_3$ incident class counts (taxonomy) with time-to-mitigation
  $CO_4$ red-team finding remediation rate
  $CO_5$ post-incident public audit pack release

## Construct validity: "what counts" as a collapse event

A CO "fires" when its pre-declared gate condition is met. To ensure validity:

- **Operationalization.** Define variable, unit, window, and trigger function.
- **Attribution.** Show that observed movement plausibly updates the policy choice (not mere noise).
- **Counter-gaming checks.** Include audits for data suppression, threshold tuning, or selection bias.
- **Error bars.** Report uncertainty and how it affects policy gating (e.g., conservative vs. optimistic collapse).

**CO template (one row per observable):**

```
CO-ID: CLM-CO3
Definition: "Realized sectoral abatement vs. plan, audited."
Window: Quarterly
Gate: collapse to Path B if realized < 70% of plan for 2 consecutive quarters.
Data: Independent verifier; public ledger link.
Anti-gaming: Randomized facility audits; penalties disclosed.
Uncertainty: ±5% measurement error; conservative gating.
```

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# The Civic Reflexivity Index (CRI)

## Components, scoring and weights

Let the six components be scored in $[0, 1]$ using anchored rubrics:

- **sra — Self-Referential Attractor density.**
  Share of agendas that *explicitly* acknowledge reflexivity/need for rule change **and** specify the scope (payoffs, constraints, information).
  Anchors: 0.00 none; 0.33 sporadic mentions; 0.66 routine acknowledgments; 1.00 systematic with scope.
- **csc — Cross-Scale Consistency.**
  Presence and quality of micro↔meso↔macro maps with declared losses/approximations.
  Anchors: 0.00 absent; 0.33 partial; 0.66 full map or two-scale with losses; 1.00 three-scale with validation.
- **co — CO Pack Rate.**
  Proportion of major agendas with a minimal CO kit meeting all design requirements.
  Anchors: 0.00 <10%; 0.33 10–39%; 0.66 40–79%; 1.00 ≥80%.
- **dcr — Delayed-Collapse Rate.**
  Fraction of non-phase-free agendas managed with a published DC plan (candidate set, gates, exit).
  Anchors: as above.
- **gdn — Gödel-Navigator Usage.**
  Rate of undecidable-loop detection with logged resolution choice (relax/detour/rebase) and timelines.
  Anchors: 0.00 no detection; 0.33 ad-hoc; 0.66 routine with logs; 1.00 institutionalized with SLAs.
- **bhc — Black-Hole Coverage.**
  Share of known non-discussable zones declared with mitigations and exit tests.
  Anchors: 0.00 taboo/implicit; 0.33 partial; 0.66 broad coverage; 1.00 comprehensive, versioned.

**Weights.** Default $w_k = \frac{1}{6}$. Legitimate alternatives must publish a rationale (e.g., security-heavy domains up-weight **gdn**/**bhc**).

## Bands and thresholds; uncertainty reporting and confidence

- **Bands:** 0–0.33 (rhetoric), 0.34–0.66 (transitional), 0.67–1.00 (reflexive-capable).
- **Decision threshold:** suggest $\mathrm{CRI} \geq 0.60$ **and** $c \geq 0.70$ for major enactments; if not, invoke DC or GN.
- **Confidence $c$.** Combine (i) data completeness (share of agendas scored; CO audit pass rate) and (ii)

inter-rater reliability. A simple composite:

$$c = 0.5 \cdot \text{completeness} + 0.5 \cdot \text{reliability}$$

with both in $[0, 1]$.

## Estimation: rubrics, exemplars, inter-rater reliability

- **Rubrics.** Publish the anchored examples above with concrete artifacts (redacted minutes, DC plans, GN logs).
- **Exemplars.** For each component, keep 3–5 gold-standard dossiers.
- **Double-scoring.** Two independent raters per agenda; report agreement.
- **Metrics.** Use Cohen's $\kappa$ (categorical anchors) or ICC (continuous scores); convert to $[0, 1]$ scale for $c$.
- **Adjudication.** Disagreements above a pre-set gap (e.g., ≥0.3) trigger review.
- **Versioning.** Keep a changelog when rubrics evolve; re-score samples to preserve comparability.

### CRI scoring worksheet (skeletal):

```
Agenda: _______   Period: _______
sra: __  evidence: link
csc: __  evidence: link
co : __  evidence: link
dcr: __  evidence: link
gdn: __  evidence: link
bhc: __  evidence: link
CRI(w): __.__   confidence c: __.__
Notes: ...
```

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# Protocols for Reflexive Governance

## Delayed Collapse (DC) for non-phase-free agendas

**When to use.** Detected path dependence (ordering effects), high irreversibility, or fragile couplings.

### DC card.

```
Inputs: Candidate set C={P1,P2,...}, Alignment signals A={A1..Ak}, Gates G with thresholds, Review cadence
T.


Steps:
1) Declare C and why each remains plausible.
2) Publish A and how each maps to COs and scales (losses disclosed).
3) Set G (e.g., "collapse to P2 if A3 and A5 fire within 2 quarters").
4) Run: observe A; update ledger; do not prune C unless gate fired.
5) Collapse: when G met with confidence c≥τ, select Pi; publish counterfactual trace.
6) Post-mortem: log timing, evidence, and costs of delay.
Guards: timebox indecision; anti-gaming audits on A; re-open only via GN.
```

## Gödel-Navigator: relax / detour / rebase decision tree

**Detection.** Cycles in dependency/justification graphs; proofs require their own conclusions; blocked by sacred constraints.

### GN decision tree.

```
Given loop L:

- RELAX: Weaken or condition an axiom (e.g., discount rate range, partial disclosure) to admit progress.

- DETOUR: Add a covering structure (pilot, sandbox, bilateral side-letter) that bypasses the obstruction
```

```
locally.

- REBASE: Change the space (jurisdiction, metric, ownership) so the loop becomes contractible (e.g.,
treaty protocol, new ledger).


SLAs:

- Assign owner; deadline T_L; publish chosen branch + rationale.

- Escalate if unresolved at T_L: second-level review; consider rebase.
Logs:
- Keep ω(L): obstruction class, attempts, outcomes. Count toward gdn.
```

### Black-Hole Registry (BHR): declaring the currently non-discussable

Some zones cannot be openly negotiated now (security, sacred values, proprietary cores). The BHR makes this explicit and safe.

#### BHR fields (one entry per zone).

```
ID, Name, Scope, Rationale (why non-discussable now),
Risks exported, Mitigations in place,
Exit test (what observation would move it into discussable space),
Review cadence, Owner, Last review date/version
```

#### Practices.

- **No stigma.** Registration is not blame; it is risk management.
- **Sunset discipline.** Each entry gets an exit test or justification for permanence.
- **Interfaces.** For each black hole, publish what *can* be discussed safely around it (perimeter COs).
- **Ties to CRI.** Coverage increases **bhc**; successful exits raise **sra/csc/co** over time.

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### Putting it together (one-page policy)

1. Adopt A1–A4 as institutional postulates.
2. Require CO kits for major agendas; publish templates.
3. Score CRI quarterly; publish scores + confidence.
4. Run DC where ordering matters; keep a visible ledger.
5. Operate GN for loops; report relax/detour/rebase choices.
6. Maintain the BHR; review and retire entries with exit tests.

These axioms, observables, indices, and protocols do not decide *who is right*. They decide *how we will know*—and when we will stop arguing in circles long enough to govern.

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# Part III — Geometry of Negotiation

# Cross-Scale Maps: Micro ↔ Meso ↔ Macro

### What must be preserved; acceptable losses; disclosure canon

A negotiation that travels across scales must state—in advance—**what the maps preserve** and **what they discard**.

#### Preserve at minimum (Invariants $\mathcal{I}$).

1. **Continuity/balance**: inputs = outputs ± storage ± loss (conservation or accounting identities).
2. **Budget/feasibility**: proposed outlays and constraints are consistent after mapping.
3. **Rights/guardrails**: hard constraints (legal, ethical, safety) are not weakened by aggregation.
4. **Time alignment**: stated horizons and lags remain comparable (no hidden time dilation).
5. **Attribution**: which levers at micro actually implement macro commitments.

**Declare acceptable losses ($\mathcal{L}$)—explicitly.**

- **Heterogeneity loss** (averages hide tails/outliers)
- **Timing loss** (asynchronous micro timing smeared into meso windows)
- **Topology loss** (network structure compressed into scalars)
- **Privacy/secrecy loss** (necessary black-boxes; show perimeter COs)
- **Phase loss** (ordering/coupling effects muted by aggregation)

**Disclosure canon (non-negotiable fields).**

- What is preserved: list $\mathcal{I}$ with checks.
- What is lost: list $\mathcal{L}$ with mitigations.
- Back-map obligations: who does what, when, with which KPI.
- CO hooks: where each **Collapse Observable** attaches at each scale.
- Version/time: data windows, refresh cadence, and expiry.

## Templates for scale-maps that travel across regimes

Use compact, auditable cards. Two templates below: a **Scale-Map** and a **Regime-Transfer** card.

**Template 1 — Scale-Map v1 (one agenda).**

```
Agenda: <name>
Window: <t0–t1>   # time bounds
# 1) Observables & Controls by scale
micro:
  observables: [o_m1, o_m2, ...]
  controls:    [u_m1, u_m2, ...]
meso:
  observables: [o_s1, ...]
  controls:    [u_s1, ...]
macro:
  observables: [o_M1, ...]
  controls:    [u_M1, ...]
# 2) Maps (forward & back)
maps:
  m_to_s: {o_m* -> o_s*, method: <agg/encoding>, loss: [heterogeneity,timing]}
  s_to_M: {o_s* -> o_M*, method: <model/weights>, loss: [topology]}
  M_to_s: {u_M* -> u_s*, method: <quota/rule>, obligation: <owner,SLA>}
  s_to_m: {u_s* -> u_m*, method: <contract/incentive>, KPI: <name>}
# 3) Invariants & Losses
invariants: [continuity, budget, rights, time_alignment, attribution]
losses:
  heterogeneity: <mitigation or "none">
  timing: <mitigation>
  topology: <mitigation>
  privacy: <perimeter COs>
  phase: <delayed-collapse used?: true/false>
# 4) CO hooks (by scale)
COs:
  micro: [CO-id -> sensor/procedure]
```

```
  meso:  [CO-id -> audit path]
  macro: [CO-id -> public ledger]
# 5) Versioning
refresh: {cadence: <e.g., quarterly>, owner: <role>}
expiry: <date or condition>
```

**Template 2 — Regime-Transfer Card (how the map survives change).**

```
Trigger: <what change of regime/rules occurs>
Continuity Plan:
  preserved: [list of invariants that remain intact]
  re-baselined: [which metrics are redefined; mapping function; side-by-side overlap window]
  black-holes: [zones that become non-discussable; perimeter COs]
Exit Test:
  data: <which COs must fire to declare the transfer complete>
  window: <persistence W>
Governance:
  owner: <institution/role>
  review: <cadence>
```

These two cards force negotiators to **carry the measurement** across boundaries (e.g., elections, mergers, treaty updates) without hiding changes inside terminology.

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# Stability and Early-Warning

## Curvature intuition: approaching tipping without declaring victory

Most systems that matter—credit networks, deterrence postures, climate-economy couplings, AI deployment—have basins of relative stability. Think of the policy state as a point moving on a surface with wells (attractors). As the system nears a tipping point, **curvature "flattens"**:

- Shocks take longer to die out (**recovery slows**).
- Variability increases (**variance rises**).
- Successive measurements look more alike (**AR(1) drifts toward 1**).
- The state "flickers" between nearby basins (**micro-hops**).

This is **warning**, not proof. Treating precursors as certainties can create the crisis they warn about—or discredit the method when a tip does not occur.

## Minimal early-warning set (variance, recovery, flicker) and responsible use

Adopt a **minimal, domain-agnostic set**—simple to compute, hard to game alone.

**Early-Warning Triplet (EWT).**

1. **Variance** of a core observable (windowed).
2. **Recovery rate** after small, identified shocks (time back to baseline).
3. **Lag-1 autocorrelation (AR1)** on the same series.

**Optional: Flicker count** (validated switches across a declared boundary) when such boundaries exist.

**Responsible-use rules.**

- **Triangulate**: act only when ≥2 of the EWT rise together beyond domain baselines.
- **Persist**: require signals to persist over a window $W$ (e.g., multiple periods) before changing posture.
- **Localize**: tie warnings to specific **CO gates**; do not free-float alarms.
- **Disclose**: publish thresholds ex-ante and keep a ledger of hits/misses.

- **Avoid reflexivity traps**: monitor whether the publication of EWI itself changes the series (and adjust gates).

**EWI card (per series).**

```
Series: <name>    Window: <t0–t1>
Baseline: {var: __, AR1: __, recovery: __}
Current:  {var: __, AR1: __, recovery: __}
Flicker:  {count: __, boundary_def: <link>}
Gate: "Warn if ≥2 of {var↑,AR1↑,recovery↓} breach baseline by >kσ for ≥W"
Ledger: [timestamp, status, action/inaction rationale]
```

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# Integrity & Anti-Gaming

## Metric-gaming archetypes and tests

Where there are thresholds, there will be gaming. Name it, test it.

**Archetypes.**

1. **Threshold tuning**: moving cutoffs to avoid firing.
2. **Selection bias**: hiding bad subpopulations or time windows.
3. **Proxy swapping**: replacing a hard-to-move metric with a softer proxy.
4. **Window shopping**: choosing a favorable averaging window after the fact.
5. **Latency doping**: delaying data to postpone bad news.
6. **Placebo compliance**: meeting the letter without the effect (paper abatement, paper safety).
7. **Hydra proxies**: splitting one risky channel into many small ones.

**Tests (build into your process).**

- **Backcast invariance**: apply today's thresholds to yesterday's data; large "improvements" that appear only after new thresholds = red flag.
- **Split-sample stability**: metrics should behave similarly across partitions (by region, time, provider).
- **Pre-reg windows**: lock window size and placement before measurement.
- **Honeypot metrics**: include a metric that should not change; if it does, manipulation likely.
- **Latency SLA**: publish data lag; deviations trigger audits.
- **Substitution alarms**: track sudden growth of near-substitutes when a regulated metric tightens.
- **Cross-metric tension checks**: if an outcome allegedly improves while its necessary inputs don't move, investigate.

## Persistence, triangulation, adversarial audits

- **Persistence**: Require metrics and EWI to meet gates for $W$ consecutive periods before collapse or reversal.
- **Triangulation**: Corroborate with at least two independent sources or methods (e.g., admin records + sensor data).
- **Adversarial audits**: Commission **red-teams** to simulate gaming strategies and test whether your gates would catch them; publish findings and fixes.

**Anti-Gaming/Audit card.**

```
Scope: <agenda/metric>

Risks: [threshold_tuning, selection_bias, proxy_swapping, window_shopping, latency_doping, placebo, hydra]

Guards:
  backcast_invariance: true
  split_sample_stability: true
  prereg_windows: {locked: true, spec: <doc>}
  honeypot_metric: <id>
  latency_SLA: <days>
  substitution_watch: <list>
Triangulation:
  sources: [S1, S2, ...]   # independent providers/methods
Red-Team Plan:
  scripts: [G1, G2, ...]
  schedule: <cadence>
  disclosure: <public/limited>
Owner: <role>   Review: <cadence>   Last_run: <date>   Findings: <link>
```

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### One-page practice summary

1. **Map the scales** with invariants and losses; publish CO hooks and back-maps.
2. **Watch curvature** responsibly: use the EWT, demand persistence + triangulation, tie to gates.
3. **Expect gaming**: design tests in, not after; red-team on a schedule; keep public ledgers where possible.

With these geometric tools—maps that survive regime change, warnings that don't cry wolf, and audits that assume ingenuity—you turn negotiation from an exchange of eloquence into a transport of measurement.

Below I start with a compact **calc-pad** (so our four cases sit on the same formal rails), then give four **timeless case templates**. Each case follows the promised rhythm:

**identify reflexivity → publish CO kit → run protocols (DC, GN, BHR) → audit CRI.**

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# Part IV — Applications (Timeless Case Templates)

# Climate Governance as a Self-Referential Attractor

### 1) Identify reflexivity

- **Discounting & equity weights**: policy changes growth/risk → changes the metrics that justified it.
- **Technology learning**: support shifts cost curves that feed back into benefit tests.
- **Exposure accounting**: adaptation alters measured damages, moving the goalposts.

### 2) CO kit (illustrative)

- **CLM-CO1** Tail exceedance: cumulative ppm-years above threshold bands; gate: escalate adaptation track if >X ppm-years in 24 months.
- **CLM-CO2** Adaptation loss index: audited service-level failures (water/health/transport); gate: if >Y events/Q for 2 quarters, reallocate ≥Z% budget.
- **CLM-CO3** Realized abatement vs plan (audited tons); gate: if <70% for 2 consecutive quarters, pivot away from price-only path.
- **CLM-CO4** Tech learning realization: LCOE/LCOH tracked vs forecast; gate: if cost curve falls ≥α% faster, advance phase-in by β months.

- **CLM-CO5** Equity disclosure: distribution of residual risk by decile/region; gate: if top-risk deciles >γ exposure, trigger compensatory transfers.
  *Anti-gaming*: randomized facility audits; back-cast invariance on thresholds; latency SLAs.

### 3) Run protocols

**Delayed Collapse (DC).**
$C$ = { P1: price-ramp, P2: sectoral standards, P3: cap-and-dividend, P4: price+standards+adaptation fast track }.
Alignment $A$ = {A1: CLM-CO3 trend, A2: CLM-CO2, A3: CLM-CO4, A4: just transition (CLM-CO5) }.
Gates $G$: e.g., *collapse to P4* if A1 falls below 0.7 for 2Q **and** A2 breaches Y in any region.

**Gödel-Navigator (GN).**
Loop L1: "right" discount rate. → **Relax**: operate with a range + scenario dominance.
Loop L2: intergenerational rights v. present voters. → **Detour**: carbon-club pilot with youth-repr council.
Loop L3: equity weights across borders. → **Rebase**: treaty side-ledger for transfers.

**Black-Hole Registry (BHR).**
Ω1: geo-engineering readiness (security). *Risks*: moral hazard; *Mitigation*: perimeter CO: incident-reporting & transparency windows; *Exit test*: independent monitoring capability online.

### 4) Audit CRI (example rubric)

- sra: 0.66 (explicit reflexivity in discounting/learning)
- csc: 0.66 (household→sector→national maps with declared losses)
- co: 0.80 (≥80% agendas with kits)
- dcr: 0.66 (DC used on non-phase-free tracks)
- gdn: 0.66 (loops logged with decisions)
- bhc: 0.66 (registry live; exit tests set)
  **CRI ≈ 0.68 (c=0.75)** → reflexive-capable; proceed with P4 under published gates.

---

## Deterrence & Arms Stability

### 1) Identify reflexivity

- **Signaling** aims to change the opponent's beliefs about rules of response—evaluations shift because signals succeed.
- **Posture changes** alter verification regimes that certify stability.
- **Escalation ladders** define the metrics they are supposed to optimize.

### 2) CO kit (illustrative)

- **DET-CO1** Alert-posture transitions (verified counts; windowed). Gate: if transitions >k/month without notification, raise stability review.
- **DET-CO2** Exercise-to-notification ratio. Gate: if <r for W months, require confidence-building measures.
- **DET-CO3** Escalation gate compliance (declared ladder vs observed behavior). Gate: any breach → convene joint incident board.
- **DET-CO4** Near-miss taxonomy counts (air/sea/cyber). Gate: rate ↑ beyond baseline by >σ for W → posture freeze.
- **DET-CO5** Hot-line latency distributions during crises. Gate: if median latency >T, mandatory comms

drill.
*Anti-gaming*: third-party sensors; cross-reporting; honeypot metrics (dummy non-sensitive drills).

### 3) Run protocols

**Delayed Collapse (DC).**
$C$ = { D1: de-alert slice, D2: notification-first doctrine, D3: reciprocal hot-line drills + ladder codification, D4: treaty-level cap on specific systems }.
Gates: collapse to D3 if DET-CO2T for 2 cycles; to D4 if DET-CO4 persists >W and verification pilots pass.

**Gödel-Navigator (GN).**
L1: "Resolve" as unverifiable essence. → **Relax**: operationalize via DET-CO3 compliance indices.
L2: Secrecy v. transparency. → **Detour**: mutually agreed *confidential COs* with escrow verification.
L3: Asymmetric capabilities. → **Rebase**: new metric space (effects-based caps instead of platform caps).

**Black-Hole Registry (BHR).**
Ω1: covert sensors; Ω2: specific warhead details. Mitigations: perimeter COs (near-miss counts, latency) + exit tests (post-program declassification windows).

### 4) Audit CRI (example)

sra 0.66, csc 0.66, co 0.66, dcr 0.66, gdn 0.66, bhc 0.80 → **CRI ≈ 0.68 (c=0.72)**. Proceed with D3; keep D4 as DC candidate.

---

## Financial Contagion & Cross-Border Rules

### 1) Identify reflexivity

- **Backstops** and resolution rules reshape "market discipline" that judges them.
- **Liquidity facilities** alter spreads and volumes used to assess success.
- **Cross-border** actions change the very legal bases that determine creditor hierarchy.

### 2) CO kit (illustrative)

- **FIN-CO1** Liquidity stress breaches (pre-declared metrics: LCR/NSFR windows). Gate: facility on/off by rule.
- **FIN-CO2** Resolution waterfall execution (who absorbed losses, when). Gate: if not executed within Δt, governance escalation.
- **FIN-CO3** Interbank spread normalization (Δ spread vs baseline). Gate: persistently >σ → expand collateral set.
- **FIN-CO4** Circuit-breaker trigger/reset logs. Gate: >n triggers/week → review microstructure.
- **FIN-CO5** Cross-border claims netting success rate. Gate: <p% → invoke treaty fallback.
  *Anti-gaming*: split-sample stability by bank classes; latency SLA on supervisory data; substitution watch for off-balance channels.

### 3) Run protocols

**Delayed Collapse (DC).**
$C$ = { F1: liquidity-only lines, F2: targeted solvency backstops, F3: bail-in first with contingent lines, F4: joint resolution fund for cross-border }.
Gates: collapse to F3 if FIN-CO1 fires repeatedly while FIN-CO2 is on time; to F4 if FIN-CO5 < p% for 2 cycles.

**Gödel-Navigator (GN).**
L1: "Too big to fail" definition. → **Relax**: size + network centrality composite with published bands.

L2: Sovereign/bank doom loop. → **Detour**: ring-fenced pilot with capped sovereign exposures.
L3: Legal conflict of resolution regimes. → **Rebase**: treaty-level *single-point of entry* clause.

**Black-Hole Registry (BHR).**
Ω1: supervisory secrets; Ω2: live-crisis liquidity taps. Mitigations: ex-post anonymized audit packs; exit tests: fixed declassification horizon.

### 4) Audit CRI (example)

sra 0.66, csc 0.66, co 0.80, dcr 0.66, gdn 0.66, bhc 0.66 → **CRI ≈ 0.68 (c=0.78)**. DC active; current collapse toward F3 with F4 on watch.

---

## AI Alignment & Safety

### 1) Identify reflexivity

- **Oversight capacity** is the very object of policy; thresholds judged by institutions the policy seeks to build.
- **Deployment rules** change evaluation telemetry (incident reporting, red-team access).
- **Capability evals** move behavior under test (training adapts to the test).

### 2) CO kit (illustrative; governance-first)

- **AI-CO1** Capability ceilings: pre-committed eval suites with hold-outs; gate: deployment blocked if >τ on restricted tasks.
- **AI-CO2** Deployment stage-gates passed/failed (pre-prod → prod); gate: rollback if gate failed + incident class ≥k.
- **AI-CO3** Incident taxonomy counts & time-to-mitigation; gate: sustained rise → reduce exposure class.
- **AI-CO4** Red-team remediation rate (findings closed/quarter); gate: <p% → freeze feature class.
- **AI-CO5** Post-incident public audit packs released within SLA; gate: miss → downgrade trust tier. *Anti-gaming*: eval rotation & secrecy windows; external auditors; proxy-swapping alarms (capability vs risk).

### 3) Run protocols

**Delayed Collapse (DC).**
$C$ = { A1: pre-deployment licensing, A2: staged deployment with kill-switch, A3: usage-based gating (capability × domain), A4: pause triggers on incident density }.
Gates: collapse to A2 if AI-CO3 rises while AI-CO1 near τ; to A4 if AI-CO5 misses SLA twice and AI-CO3 escalates.

**Gödel-Navigator (GN).**
L1: "AGI" definition loop. → **Relax**: operate on capability ceilings & hazard classes, not labels.
L2: Proprietary weights v. auditability. → **Detour**: *confidential COs* under NDA with public perimeters.
L3: Jurisdictional mismatch. → **Rebase**: cross-border registry + recognition protocol.

**Black-Hole Registry (BHR).**
Ω1: model weights; Ω2: red-team exploit details. Mitigations: perimeters via AI-CO3/4/5; exit tests: time-delayed disclosure or capability decay.

### 4) Audit CRI (example)

sra 0.80 (explicit), csc 0.66, co 0.80, dcr 0.66, gdn 0.66, bhc 0.66 → **CRI ≈ 0.71 (c=0.74)**. Proceed with A2 under published gates; keep A4 as DC fallback.

---

# Timeless use note

These templates are **frame-agnostic**. They do not pick winners; they force **measurement that travels**. If you reuse them a century from now, swap in contemporary observables but keep the discipline: declare reflexivity, pre-register COs, run DC/GN/BHR, and publish a CRI with confidence.

---

# Part V — Philosophy and Limits

# Where This Sits Among Theories

## Realism, liberalism, constructivism vs. axiomatic reflexivity

**Axiomatic reflexivity** is not a rival grand theory of outcomes; it is a **discipline for how any theory makes its claims discussable** when the rules are themselves at issue.

- **Realism (power and security first).**
  Reflexivity agrees that capability distributions and credible threats matter. It asks realists to **pre-register COs** for stability (e.g., near-miss counts, alert-posture gates) and to run **Delayed Collapse** where order effects are fatal. Power is not denied; it is **measured across scales** and constrained by audit trails that keep signaling from collapsing into theater.

- **Liberalism (institutions and interdependence).**
  Institutions are precisely where reflexive protocols live: **Gödel-Navigator** becomes the rulebook for revising rules; the **Black-Hole Registry** expresses what a regime cannot yet publicize and how it mitigates that opacity. Liberalism supplies the venues; reflexivity supplies the **measurement contracts** that keep cooperation falsifiable.

- **Constructivism (norms constitute interests).**
  If language and identity shape interests, then governance must **measure norm stabilization**: COs can be counts of rule adoption, compliance narratives, or de-escalatory speech acts with observable downstream effects. Reflexivity does not deny construction; it insists that construction produce **observables that travel**.

The upshot: axiomatic reflexivity sits **orthogonally** to these traditions. It neither predicts the world nor prescribes its values; it **binds** any worldview to a regime of observables, gates, and audit that survive regime change and rhetorical skill.

## Normative neutrality and the ethics of disclosure

The **Civic Reflexivity Index (CRI)** and its protocols are **normatively thin**: they do not pick ends (security vs welfare vs liberty). They do impose **ethical constraints on the means**—chiefly around disclosure:

**Disclosure Ethics Canon (operational).**

1. **Proportionality.** Disclose enough to make claims discussable; withhold only what, if revealed, would cause disproportionate harm relative to the gain in accountability.

2. **Perimeterization.** When cores must remain dark (security, privacy, IP), publish **perimeter COs** so the public can still track effects.

3. **Reciprocity.** Where verification creates unilateral exposure, negotiate **reciprocal COs** or escrowed/confidential COs with independent auditors.

4. **Sunset & Review.** Every secrecy claim gets a sunset date or an **exit test**; schedulers must prove continuation if extended.

5. **Non-maleficence.** Redact micro-identifiers that would harm vulnerable populations while preserving aggregate CO value.

6. **Epistemic humility.** Publish **uncertainty** and **confidence** with the same prominence as the point estimate.

Neutrality here is not indifference. It is the decision to **make measurement a first-class civic act**, and to treat secrecy as an addressable engineering constraint rather than a conversation-stopper.

---

# What CRI Cannot Do

## Undecidability remains; trade-offs and tragic choices

CRI acknowledges **undecidable loops** (A3). It can *route* them:

- **Relax** a premise (broaden discount-rate bands),
- **Detour** to a cover (pilots/sandboxes),
- **Rebase** the domain (new jurisdiction/metric).

But it cannot **dissolve** real conflicts of value. Some choices are tragic because goods are **incommensurable** (e.g., security vs due process when time is short). CRI's role is to **name the incomparability**, show the **loss record** (who bears what), and keep the ledger open for later repair.

## Failure modes: ritualization, stagnation, false precision

Name them early; design against them.

**1) Ritualization (Goodhart's trap).**
When a CO becomes a target, it ceases to be a good measure. Symptoms: checklists pass while the world worsens; "paper compliance" spikes.

- *Guards.* Rotate COs; run **backcast invariance**; attach **honeypot metrics**; publish **cross-metric tension checks** (Part III).

**2) Stagnation (permanent Delayed Collapse).**
DC without closure becomes abdication. Symptoms: ever-expanding candidate sets; moving gates; "review next quarter" for years.

- *Guards.* **Closure SLAs**: collapse unless two explicit conditions justify extension; mandatory **counterfactual trace** when collapsing; timebox indecision and escalate.

**3) False precision (spurious certainty).**
A glossy CRI at 0.71 with **confidence** $c = 0.34$ is not governance, it's numerology.

- *Guards.* Publish $c$ alongside CRI; require **triangulation**; if reliability drops below threshold, **freeze enactment** or widen bands.

**4) Capture and performativity.**
Metrics chosen by the most powerful; disclosure theater replaces accountability.

- *Guards.* **Blind rescoring** by independent raters; **weight-justification ledger** (any weight change must cite evidence); periodic **red-team audits**; minimum participation of affected parties in rubric updates.

---

# Practice Cards (reusable)

### A. Meta-Audit for CRI (self-check).

```
Scope: CRI process (period __)
Completeness: __.__   Reliability: __.__   Confidence c: __.__
Blind Rescore: done? [Y/N]  delta: __.__
Weights Updated? [Y/N]  Rationale link: <url>
```

```
Red-Team Findings: <summary>  Fixes applied: <summary>
Closure SLA breaches: count=__  Justifications logged: <links>
Decision: enact | defer | re-open GN | revise COs
```

### B. Tragic Choice Register (when values collide).

```
Agenda: <name>   Collision: <values at stake>
COs affected: [ ... ]   Groups affected: [ ... ]
Loss Record: who bears residual risk, how measured
Mitigations: near-term | long-term
Revisit date: <date>   Exit tests: <observations>
```

### C. Capture/Performativity Test.

```
Symptoms: [elite-only rubric edits, sudden CRI jumps post-weight change, PR-first disclosures]
Diagnostics: split-sample stability? [Y/N]  backcast invariance? [Y/N]
Remedies: rotate COs, add external raters, lock windows, publish raw series
```

---

## A closing note on philosophy

Reflexivity is an **ethic of form**. It does not replace the human work of caring about ends; it insists that when we disagree about ends—and worse, when the rules are part of the dispute—we **move the argument into a space that remembers**: observables that travel, gates that do not shift, logs that a stranger can audit a century later.

That is not a substitute for wisdom. It is a scaffold strong enough for wisdom to climb.

---

# Part VI — Institution Design

# The CRI Dashboard

## Public artifacts: CO packs, scale-maps, loop logs, black-hole ledger

Make a single public surface where citizens, partners, and future auditors can *see the same thing*. Minimal files:

```
# /cri/dashboard/index.yml
period: 2025Q3
agendas: [Climate, Deterrence, Finance, AI]
artifacts:
  - co_packs:      [/cri/co/Climate.yml, /cri/co/AI.yml, ...]
  - scale_maps:    [/cri/scalemaps/Climate.yml, ...]
  - loop_logs:     [/cri/gn/loops.csv]
  - black_holes:   [/cri/bhr/registry.csv]
  - dc_plans:      [/cri/dc/Climate.yml, /cri/dc/AI.yml]
  - ewi_panels:    [/cri/ewi/Climate.csv, ...]
  - cri_scores:    [/cri/score/CRI_timeline.csv]
  - thresholds:    [/cri/policy/thresholds.yml]
version: v1.2
owner: Secretariat for Reflexive Governance
```

### CO pack (per agenda).

```
# /cri/co/AI.yml
agenda: AI
cos:
  - id: AI-CO1
    def: "Capability ceiling suite"
    window: quarterly
    gate: "block deploy if >τ on restricted tasks"
    data: "external auditor + hold-out evals"
    anti_gaming: ["eval rotation","secrecy window"]
# 4–8 more rows ...
```

### Scale-map.

```
# /cri/scalemaps/Climate.yml
invariants: [continuity,budget,rights,time_alignment,attribution]
losses:
  heterogeneity: "tails preserved via percentile COs"
  topology: "network aggregated; mitigation: stress COs on hubs"
  privacy: "facility IDs redacted; perimeter COs published"
maps:
  m_to_s: {method: "weighted agg", loss: [heterogeneity,timing]}
  s_to_M: {method: "sector model", loss: [topology]}
  M_to_s: {obligation: "quota rule", owner: "Ministry X"}
CO_hooks:
  micro: [CLM-CO3]
  meso:  [CLM-CO2]
  macro: [CLM-CO1,CLM-CO5]
refresh: {cadence: quarterly, owner: "Climate Unit"}
```

### Gödel-Navigator loop log.

```
# /cri/gn/loops.csv

loop_id,agenda,class,detected,owner,decision,deadline,status,notes

L-001,Climate,discount-rate,2025-06-12,MoF,RELAX,2025-09-30,open,"band [1.5,3.5]%, scenario dominance"

L-014,Deterrence,secrecy-vs-verification,2025-07-02,MOD,DETOUR,2025-10-01,closed,"confidential CO escrow"
```

### Black-Hole Registry.

```
# /cri/bhr/registry.csv

id,name,scope,risks,mitigations,exit_test,review_cadence,owner,last_review

Ω-001,Geoengineering readiness,dual-use R&D,"moral hazard","perimeter incident COs","independent monitors
online",semiannual,Science Council,2025-06-30

Ω-007,Model weights,AI foundation models,"IP/security","AI-CO3/4/5 perimeter","time-delayed
disclosure",quarterly,Digital Regulator,2025-07-15
```

### DC plan.

```
# /cri/dc/AI.yml
candidates: [A1: licensing, A2: staged+kill-switch, A3: usage-based gating]
alignment_signals: [AI-CO1, AI-CO3, AI-CO4, AI-CO5]
gates:
  - "collapse→A2 if AI-CO3↑ and AI-CO5 misses 2 SLAs"
  - "reopen if c<0.6 for 2 cycles"
ledger: /cri/dc/AI_ledger.csv
timebox: "max 3 quarters before collapse or escalate GN"
```

### CRI timeline.

```
# /cri/score/CRI_timeline.csv
period,sra,csc,co,dcr,gdn,bhc,weighting,CRI,confidence
2025Q1,0.58,0.52,0.60,0.40,0.41,0.50,eq,0.50,0.66
2025Q2,0.62,0.58,0.68,0.55,0.50,0.56,eq,0.58,0.71
2025Q3,0.66,0.62,0.75,0.60,0.58,0.62,eq,0.64,0.74
```

## Thresholds for enactment; exception handling

### Default enactment policy.

```
# /cri/policy/thresholds.yml
enact_if:
  CRI: ">= 0.60"
  confidence: ">= 0.70"
  component_floors:
    co:  ">= 0.60"
    gdn: ">= 0.50"
    bhc: ">= 0.50"
else:
  path: "Run DC or GN; publish plan & ledger"
```

**Waiver Ledger (for rare exceptions).**

```
# /cri/policy/waiver.yml
id: W-2025-03
agenda: Deterrence
justification: "Imminent risk window requires interim posture before CRI gates"
mitigations: ["higher EWI cadence","external observers","shorter horizon"]
sunset: "2026-01-01"
audit_intensification: "monthly"
sign_off: {dual_key: ["Council Chair","Opposition Lead"], votes: "≥2/3"}
counterfactual_trace: "/cri/waiver/W-2025-03_trace.pdf"
```

No silent exceptions: every waiver is time-boxed, counterfactually documented, and logged.

---

# Treaties, Councils, and Reviews

## Embedding Delayed-Collapse and Gödel-Navigator in procedures

**Treaty/Council Charter Language (model).**

> **Article X — Reflexive Procedures.**
> (1) *Delayed Collapse*. For non-phase-free agendas, the Council shall maintain a candidate set $C$, publish alignment signals $A$, and pre-declare gates $G$. A **Collapse Vote** may be scheduled *only* upon certification that a gate has fired; minutes must include the evidence ledger and counterfactuals.
> (2) *Gödel-Navigator*. The Secretariat shall keep a **Loop Docket**. For any undecidable loop $L$, the Council must, within the SLA, select **Relax**, **Detour**, or **Rebase**, and publish a rationale. Unresolved $L$ at deadline escalates to an extraordinary session.
> (3) *Black-Hole Registry*. Non-discussable zones shall be recorded with risks, mitigations, and exit tests; each entry must carry a review cadence and owner.

**Loop Docket (operating rule).**

- Intake: any member may file $L$; Secretariat triages class (discounts, secrecy, jurisdiction, sacred).
- SLA: owner named; decision within $T_L$ (e.g., 60 days).
- Hearing: Measurement Referee verifies evidence; Protocol Marshal enforces options scope.
- Output: decision + logging to `/cri/gn/loops.csv`.

**Collapse Vote choreography.**

- Gate verification by Measurement Referee.
- Presentation of counterfactual trace (what would have happened under other candidates).
- Vote on collapse **to the gate-qualified candidate only**; no opportunistic substitution.

## Periodic review and weight updates (published rationale)

**Review cadence.**

- **Quarterly:** CRI update, artifacts refresh, EWI panels.
- **Semiannual:** BHR review; exit-test decisions.
- **Annual:** Rubric & weight review with *backcast invariance* test.

**Weight update protocol.**

```
# /cri/policy/weights_2026.yml
w: {sra: 0.20, csc: 0.20, co: 0.20, dcr: 0.15, gdn: 0.15, bhc: 0.10}
rationale: "Higher emphasis on measurement kits & scale integrity after 2025 findings"
```

```
backcast_test:
  delta_mean: "+0.02 CRI across 2019–2024"
  anomalies: ["2021Q4 Finance spike investigated; resolved by re-scoring co with new rubric"]
publication: "2025-12-15"
```

Every change ships with side-by-side old/new CRI series, anomalies explained, and a public comment window.

---

# Education for Reflexivity

## Diplomatic drills; referee roles; red-team protocols

### Reflexivity Roles (embedded in meetings).

- **Measurement Referee.** Certifies CO definitions, gate hits, scale-map invariants; runs backcast and split-sample checks.
- **Protocol Marshal.** Enforces DC/GN/BHR procedures, timeboxes rhetoric, triggers Collapse Votes only on gate events.
- **Ledger Custodian.** Maintains artifacts, publishes updates, ensures versioning and audit trails.

### Core drills (90–120 minutes each).

1. **CO-Capture Drill.** Teams take a live agenda; produce a 5–9 item CO kit with anti-gaming checks; peer review via honeypot metrics.
2. **Scale-Map Drill.** Build a micro↔meso↔macro map; list $\mathcal{I}$, $\mathcal{L}$, back-maps; present perimeter COs for any privacy losses.
3. **EWI Triage Drill.** Given noisy series, compute variance/AR1/recovery; set gates; decide warn/no-warn under persistence + triangulation.
4. **DC Closure Drill.** Run a candidate set $C$ with simulated signals $A$; decide collapse under pre-declared gates; write the counterfactual trace.
5. **GN Loop Handling.** Classify loops; choose relax/detour/rebase; draft the docket entry with SLA and owner.
6. **BHR Perimeterization.** Turn a sensitive core into a safe perimeter: define public COs, mitigations, and exit tests.

### Red-team protocol (semiannual).

```
# /cri/redteam/plan.yml

scope: ["Climate co-pack thresholds","Deterrence latency CO","Finance resolution logs","AI incident
taxonomy"]

scripts:
  - "Threshold tuning under budget pressure"
  - "Latency doping via reporting chains"
  - "Proxy swapping (capability→benign proxy)"
  - "Window shopping (after-the-fact averaging)"
cadence: "twice yearly"
disclosure: "public summary; sensitive details under NDA"
owner: "Independent Audit Panel"
fix_tracking: "/cri/redteam/fixes.csv"
```

## From debate to discussability

Rewire the meeting template so **measurement building precedes persuasion**:

1. **Reflexivity statement** (is the agenda self-referential?)

2. **CO kit** (draft live; agree on gates)
3. **Scale-map** (declare invariants/losses; assign back-maps)
4. **Protocol assignment** (DC? GN? BHR entries?)
5. **Only then** open value-arguing within the measurement frame.

**Meeting scaffold (one page).**

```
# /cri/process/meeting.yml
agenda: <name>  chair: <name>
segments:
  - 15m Reflexivity + Scope
  - 30m CO Kit Build (Referee chairs)
  - 20m Scale-Map + Back-Maps
  - 10m Protocol Assignment (DC/GN/BHR)
  - 30m Debate within frame
  - 15m Decisions & Artifact Publishing
outputs:
  - /cri/co/<agenda>.yml
  - /cri/scalemaps/<agenda>.yml
  - /cri/gn/loops.csv (if any)
  - /cri/dc/<agenda>.yml (if DC)
```

---

### What this buys you

- **Continuity across regimes.** Artifacts survive elections and crises.
- **Auditability over time.** A stranger in 2125 can reconstruct what you knew, measured, and promised.
- **Civic legitimacy.** Disagreement remains, but *discussability* is guaranteed: gates don't shift, logs persist, exceptions sunset, and red-teams keep everyone honest.

With a dashboard, chartered procedures, and a living pedagogy, **reflexivity stops being a slogan** and becomes something your institutions *do*.

---

# Summary

This article builds a procedure-first architecture for governing problems that defeat ordinary argument. When a proposal would change the very rules used to judge it, debate becomes a **rhetorical trap**: both sides can be right inside their own frames, yet nothing converges. We recast civilization not as the sum of artifacts but as a capacity for **reflexive governance**: the ability to make such problems discussable.

The foundation is four axioms. **A1 (Reflexivity)** names the circularity. **A2 (Observability)** requires every serious claim to bind at least one shared **Collapse Observable (CO)**—a pre-declared, auditable event or threshold that narrows disagreement. **A3 (Incompleteness)** legitimizes structured rule-change when loops block progress. **A4 (Delayed-Collapse)** keeps multiple options live in non-phase-free regimes until alignment signals cross gates. These are not ideologies; they are reusable contracts for measurement and decision.

From these, we construct the **Civic Reflexivity Index (CRI)**—a six-part score of a polity's reflexive capacity: explicit recognition of reflexivity (**sra**), cross-scale consistency (**csc**), coverage of valid CO kits (**co**), disciplined use of **Delayed-Collapse** (**dcr**), handling of undecidable loops via the **Gödel-Navigator** (**gdn**), and accountable management of the currently non-discussable via a **Black-Hole Registry** (**bhc**). CRI is always published with **confidence**—a composite of completeness and inter-rater reliability—because precision without reliability is theater.

Negotiation gets a **geometry**. We require micro↔meso↔macro **scale-maps** that state what is preserved (**invariants**) and what is lost (heterogeneity, timing, topology, privacy, phase), with mitigations and back-map obligations. We use a minimal early-warning triplet—variance, recovery, and AR(1)—only with persistence and triangulation, and only when tied to CO gates, to avoid crying wolf.

Three **protocol cards** operationalize the axioms. **Delayed-Collapse** defines candidate sets, alignment signals, gates, timeboxes, and counterfactual traces at the moment of collapse. The **Gödel-Navigator** routes loops by **relaxing** axioms, **detouring** through pilots or escrowed observables, or **rebasing** the metric/venue. The **Black-Hole Registry** acknowledges necessary opacity without surrendering accountability by publishing risks, mitigations, perimeter COs, and exit tests with review cadences.

Part VI shows how to make the system durable. A public **CRI Dashboard** aggregates CO packs, scale-maps, loop logs, black-hole ledgers, DC plans, and CRI time series with thresholds and waiver records. Treaty and council language embeds DC and GN as procedure—not afterthoughts—and requires audits, counterfactual traces, and closure SLAs for decisions. An education kit institutionalizes drills (CO capture, scale-mapping, EWI triage, DC closure, GN routing, BHR perimeterization) and assigns roles: **Measurement Referee**, **Protocol Marshal**, and **Ledger Custodian**.

Limits are explicit. CRI does not choose ends or dissolve tragic value conflicts; it ensures that when ends collide, society owns the losses in a ledger that a stranger can audit a century later. Failure modes—ritualization, stagnation, false precision, capture—are anticipated with anti-gaming tests, red-team audits, and weight-change backcasts.

What remains, finally, is a shift in civic habit: from **debate first** to **measurement first**. By binding claims to observables, mapping scales with declared losses, running disciplined protocols, and publishing auditable artifacts, polities can move hard problems from eloquence to evidence—turning rhetoric into governance that travels across regimes and time.

---

# Appendix A — Scoring Rubrics & Worked Examples

# A1. Component Rubrics (anchored)

For each component, select the level that **best matches the evidence**; if between anchors, interpolate (e.g., 0.50).

### sra — Self-Referential Attractor Density

**Definition:** Share of agendas that explicitly acknowledge reflexivity (what rules/metrics their adoption would change) **and** scope it (payoffs/constraints/information).

- **0.00** No reflexivity statements; proposals argued as if rules were fixed.
- **0.33** Occasional acknowledgments without scope or artifacts.
- **0.66** Routine, agenda-by-agenda statements; each links to a Reflexivity note (1–2 pages).
- **1.00** Systematic: every major agenda has scoped reflexivity + integration into procedures (DC/GN/BHR references).

**Evidence pack:** reflexivity notes, council minutes citing them.

---

### csc — Cross-Scale Consistency

**Definition:** Quality of micro↔meso↔macro **scale-maps** with declared invariants $\mathcal{I}$ and losses $\mathcal{L}$, plus back-map obligations.

- **0.00** No maps; siloed reasoning.
- **0.33** Partial maps (two scales) or missing losses; back-maps unclear.
- **0.66** Three-scale map with $\mathcal{I}$, $\mathcal{L}$, back-maps (owners & SLAs) published.
- **1.00** As 0.66 **plus** validation runs (spot checks that back-maps executed; perimeter COs for privacy).

**Evidence pack:** scale-map YAML, implementation SLAs, validation memos.

---

## co — CO Pack Rate

**Definition:** Proportion of major agendas carrying a **CO kit** (5–9 observables) that passes design checks (pre-reg, auditability, frame-bridging, anti-gaming, cross-scale hooks).

- **0.00** < 10% agendas with CO kits.
- **0.33** 10–39%.
- **0.66** 40–79%.
- **1.00** ≥ 80% (and ≥ 1 independent audit passed per quarter).

**Evidence pack:** CO kit files, audit confirmations.

---

## dcr — Delayed-Collapse Rate

**Definition:** Fraction of non-phase-free agendas running **Delayed-Collapse** (candidate set $C$, alignment signals $A$, gates $G$, ledger).

- **0.00** DC not used; early lock-in normal.
- **0.33** Ad-hoc DC on select cases; gates vague.
- **0.66** DC standard for path-dependent agendas; gates pre-declared; counterfactual traces published at collapse.
- **1.00** DC institutionalized with timeboxes, reopen rules, and post-mortems.

**Evidence pack:** DC plans, collapse ledgers, counterfactual traces.

---

## gdn — Gödel-Navigator Usage

**Definition:** Rate and quality of **loop detection** and routing (relax/detour/rebase) with SLAs and logs.

- **0.00** Loops unrecognized; debates stall.
- **0.33** Some loops named; decisions not logged.
- **0.66** Loop docket with classes, owners, SLA compliance; decisions & rationales published.
- **1.00** Same as 0.66 + escalation working; periodic synthesis of loop classes and fixes.

**Evidence pack:** loop log CSV, decisions, SLA dashboard.

---

## bhc — Black-Hole Coverage

**Definition:** Coverage and quality of **Black-Hole Registry** (non-discussables) with risk notes, mitigations, **exit tests**, and review cadence.

- **0.00** Sensitive zones taboo/implicit.
- **0.33** Partial list; few mitigations; no exits.
- **0.66** Comprehensive list with mitigations & exit tests; reviews on schedule.
- **1.00** As 0.66 + documented exits (retired entries) and perimeter COs live.

**Evidence pack:** BHR CSV, review minutes, exit records.

---

# A2. Scoring Worksheet (one agenda)

```
agenda: <name>              period: <YYYY-Q#>
scores:
  sra: 0.__   evidence: <link(s)>
  csc: 0.__   evidence: <link(s)>
  co:  0.__   evidence: <link(s)>
  dcr: 0.__   evidence: <link(s)>
  gdn: 0.__   evidence: <link(s)>
  bhc: 0.__   evidence: <link(s)>
weights: {sra: 0.1667, csc: 0.1667, co: 0.1667, dcr: 0.1667, gdn: 0.1667, bhc: 0.1667}
CRI: 0.__    confidence: 0.__
completeness: 0.__   reliability: 0.__ (method: kappa|ICC; raters: N)
notes: |
  - key gaps, planned fixes
  - any waiver in effect? link
```

**Portfolio roll-up:** average (weighted by agenda materiality if defined) across agendas; publish component breakdown and time series.

---

# A3. Estimation & Audit Playbook

**Raters & reliability.**

- Two independent raters per agenda; rotate every 2–3 cycles.
- Use Cohen's $\kappa$ for anchored categories; rescale to $[0, 1]$ as reliability.
- If $\kappa < 0.6$, schedule rubric calibration; publish deltas.

**Backcast invariance.**

- Recompute last 8 quarters with current rubric/weights.
- Flag anomalies: changes > 0.08 not tied to new evidence or genuine reforms.
- Publish a short "delta note" per anomaly.

**Blind re-score & adjudication.**

- If any component differs by ≥ 0.30 between raters, assign a senior adjudicator; document rationale.

**Waiver handling.**

- Waivers **do not** directly alter CRI; they change the **decision rule** (e.g., allow enactment under lower $c$ with added mitigations).
- Every waiver must have sunset + counterfactual trace plan.

---

# A4. Worked Example 1 — Climate (2025Q3)

**Artifacts (public):**

- CO pack: `/cri/co/Climate.yml` (CLM-CO1..CO5)
- Scale-map: `/cri/scalemaps/Climate.yml`
- DC plan: `/cri/dc/Climate.yml`
- Loop log: `L-001 discount-rate=RELAX`
- BHR: `Ω-001 geoengineering readiness`

**Scoring (anchors → score).**

- **sra 0.66** — Reflexivity statements cover discounting, tech-learning; link to L-001.
- **csc 0.66** — Three-scale map with invariants/losses; back-maps owned by ministries; validation spot checks passed.
- **co 0.80** — ≥ 80% agendas have CO kits; independent audit cleared CLM-CO3/4 methods.
- **dcr 0.66** — DC active: P1 price-ramp, P2 standards, P4 blended; gates tied to CLM-CO2/3/4.
- **gdn 0.66** — Loop docket active; L-001 relaxed to bands; SLA met.
- **bhc 0.66** — BHR lists geoengineering; perimeter COs defined; exit test = monitors online.

**Weights (equal).**
CRI = mean = $(0.66 + 0.66 + 0.80 + 0.66 + 0.66 + 0.66)/6 = 0.685 \approx$ **0.69**.

**Reliability & completeness.**

- completeness 0.86 (6/7 agendas scored; one minor agenda pending).
- reliability 0.72 ($\kappa$ on categorical anchors).
- $c = (0.86 + 0.72)/2 = 0.79$.

**Decision:** CRI 0.69, $c$ 0.79 → **meets gates**. Continue DC; collapse to P4 if CLM-CO3 < 0.70 for 2Q and CLM-CO2 breaches in any region. Publish counterfactual trace upon collapse.

**Audit notes:** backcast shows +0.03 over 2024 series due to added CO audits (documented).

---

# A5. Worked Example 2 — Deterrence (2025Q3)

**Artifacts (public):**

- CO pack: `/cri/co/Deterrence.yml` (DET-CO1..CO5)
- Loop log: `L-014 secrecy-vs-verification=DETOUR` (confidential CO escrow)
- BHR: `Ω-002 specific warhead specs; exit test=post-program declassification`
- DC plan: `/cri/dc/Deterrence.yml` (D1..D4)

**Scoring.**

- **sra 0.66** — Statements on signaling reflexivity & verification regime effects.
- **csc 0.66** — Scale-map addresses micro (unit rules) ↔ meso (theater) ↔ macro (strategic stability).
- **co 0.66** — COs defined; third-party sensors included; audit pending on DET-CO5 latency.
- **dcr 0.66** — DC for posture packages (D1..D4) with gates on DET-CO2/4/5.
- **gdn 0.66** — Loop docket used; secrecy vs verification handled via detour (escrow auditors).
- **bhc 0.80** — BHR comprehensive; several entries retired with exit tests passed.

**CRI:** $(0.66 + 0.66 + 0.66 + 0.66 + 0.66 + 0.80)/6 = 0.68$.
**Confidence:** completeness 0.78, reliability 0.70 → $c = 0.74$.
**Decision:** Meets gates. Schedule quarterly hot-line drills until DET-CO5 median < T.

**Audit notes:** blind re-score Δmax=0.17 (DET-CO3 compliance index), below adjudication threshold.

---

# A6. Gold-Standard Dossiers (for training raters)

Maintain 3–5 “exemplar” dossiers per component. Each dossier contains:

- **Artifact bundle:** CO pack, scale-map, DC plan, loop/BHR records.
- **Why it’s gold:** short memo mapping artifacts → rubric language.
- **Common pitfalls:** 2–3 examples where similar-looking evidence **doesn’t** qualify (e.g., COs without anti-gaming checks).

**Example titles:**

- *sra-G1:* “Explicit reflexivity across fiscal, tech, information rules (Climate 2024Q4)”
- *csc-G2:* “Back-map validation with SLAs (Finance 2025Q1)”
- *co-G3:* “Frame-bridging COs with independent audit (AI 2025Q2)”
- *dcr-G1:* “Counterfactual trace at collapse (AI 2025Q3)”
- *gdn-G1:* “Loop class synthesis & escalation (Deterrence 2025Q2)”
- *bhc-G2:* “Perimeter COs + exit tests (Dual-use R&D 2025)”

---

# A7. Quick Calibration Checklist (per scoring round)

- Portfolio in scope defined; agendas and materiality confirmed.
- Rubric version pinned; weight file hashed; backcast run.
- Rater A/B briefed; gold-standard dossiers reviewed.
- All six components scored **with evidence links**.
- Reliability computed; disagreements ≥ 0.30 flagged.
- CRI & *c* computed; component floors checked.
- Decision logic executed (enact | DC | GN | waiver).
- Public dashboard updated; artifacts versioned and published.
- Red-team findings (if any this period) integrated into next rubric revision plan.

---

# A8. Minimal Files (drop-in templates)

### Weights file

```
# /cri/policy/weights.yml
w: {sra: 0.1667, csc: 0.1667, co: 0.1667, dcr: 0.1667, gdn: 0.1667, bhc: 0.1667}
since: 2025-01-01
rationale: "equal weights at launch; revisit annually"
```

### Score table (machine-readable)

```
period,agenda,sra,csc,co,dcr,gdn,bhc,CRI,confidence,completeness,reliability,weights_version
2025Q3,Climate,0.66,0.66,0.80,0.66,0.66,0.66,0.69,0.79,0.86,0.72,v1
2025Q3,Deterrence,0.66,0.66,0.66,0.66,0.66,0.80,0.68,0.74,0.78,0.70,v1
```

### Adjudication note

```
agenda: Finance
```

```
component: co
raterA: 0.66   raterB: 0.33   delta: 0.33
decision: "Raise to 0.50 (interpolated): CO kit passes pre-reg & auditability; anti-gaming incomplete."
rationale: "Missing split-sample stability; improvement plan logged."
owner: "Measurement Referee"   date: 2025-08-29
```

---

**Use this appendix as the house manual:** it fixes the semantics of your scores, the evidence required, and the math you'll publish. With it, anyone—today or a century from now—can reconstruct what you claimed, why you scored it that way, and how seriously to take the number.

---

# Appendix B. Reflexive Protocol Cards (Delayed-Collapse, Gödel-Navigator, Black-Hole Registry)

# Calc-Pad

**Objects.**

- Candidate set $C = \{P_i\}$; Alignment signals $A = \{A_j\}$; Gates $G = \{g_k\}$.
- Loops $L$ (undecidable/self-referential), with actions $\{\text{RELAX}, \text{DETOUR}, \text{REBASE}\}$.
- Black-hole zones $\Omega$ (currently non-discussable), each with perimeter COs and exit tests.

**Invariants.**

- **I1. Pre-declaration**: COs, gates, SLAs exist *before* action.
- **I2. Single source of truth**: every step logged in public (or perimeter) ledgers.
- **I3. Closure discipline**: DC must either collapse or escalate via GN by the timebox; GN must pick a branch or escalate; BHR entries must have sunset or exit tests.

**Guards.**

- Persistence windows, triangulation across data sources, anti-gaming tests (backcast, split-sample, honeypots), counterfactual traces at collapse.

---

# CARD 1 — Delayed-Collapse (DC)

**Purpose.**
Safely decide in **non-phase-free** regimes (ordering matters) by **keeping ≥2 candidates live** until pre-declared alignment signals pass gates.

**Use when.**
Irreversibility, path dependence, high coupling, or governance capacity still forming.

**Inputs.**

- $C$: candidate policies $P_1 .. P_n$ (each justified as plausibly optimal ex-ante).
- $A$: alignment signals tied to **COs** (observables with methods & owners).
- $G$: gate conditions ("if $f(A) \geq \theta$ for ≥W, collapse→$P_k$").
- Timebox $T$, review cadence, owners, anti-gaming suite.

**Operating steps.**

1. **Declare** $C, A, G, T$ and publish the DC plan.

2. **Observe & log** $A$ on cadence; run persistence + triangulation checks.
3. **Do not prune** candidates except by gate or safety grounds.
4. **Collapse** only when a gate fires *and* confidence $c \geq \tau$.
5. **Publish counterfactual trace** showing how other candidates would have fared under observed $A$.
6. **Escalate** to GN if timebox $T$ expires without a gate.

**YAML stub (per agenda).**

```
# /cri/dc/<agenda>.yml
candidates: [P1, P2, P3]
alignment_signals:
  - id: A1
    co_ref: <CO-ID>
    method: <link>
    owner: <role>
gates:
  - id: G1
    rule: "collapse->P2 if {A1<th1 & A3>th3} persists >= W"
    W: "2 quarters"
confidence_threshold: 0.70
timebox: "3 quarters"
reopen_rules: "Only via GN decision"
ledger: "/cri/dc/<agenda>_ledger.csv"
```

**Ledger (CSV).**

```
t,signal_id,value,check(persist,triang),gate_hit?,note
2025-05-15,A1,0.63,TRUE/FALSE,FALSE,"..."
```

**Anti-gaming (must include).**

- Backcast invariance on gate thresholds; pre-reg windows; split-sample stability; honeypot metric.

**Common failure modes → fixes.**

- *Perma-DC*: add **closure SLA** + escalation to GN.
- *Gate creep*: lock thresholds; any change requires weight-justification ledger + backcast.

**Mini-example.**
AI deployment: $C$ = {license-first, staged+kill-switch, usage-gating}.
Gate: "collapse→staged if incident density (AI-CO3) ↑ and audit SLA (AI-CO5) misses twice within W."

---

# CARD 2 — Gödel-Navigator (GN)

**Purpose.**
Handle **undecidable/self-referential** loops without stalling: **RELAX** an axiom, **DETOUR** around it, or **REBASE** the domain.

**Loop detectors (use ≥1).**

- Circular dependency graphs ("needs itself").
- Proof requires its own conclusion (begging the rule).
- Sacred/secret constraints forbid needed evidence.
- Jurisdictional mismatch makes compliance undefined.

**Classification (examples).**

- *Discount band loops* (e.g., climate discount rate).

- *Secrecy vs verification* (deterrence telemetry).
- *Jurisdiction/metric* (cross-border bank resolution).
- *Identity/norm collision* (rights vs security).

**Decision tree.**

- **RELAX** — weaken/condition an axiom enough to admit progress (e.g., banded discount rates + dominance tests).
- **DETOUR** — add a local cover (pilot, sandbox, side-letter, escrow auditor) to create evidence without global commitment.
- **REBASE** — change metric space or venue (treaty protocol, recognition registry, single-point-of-entry).

**SLA & escalation.**
Assign owner + deadline $T_L$. Unresolved → extraordinary session; consider rebase by default.

**YAML stub (loop docket).**

```
# /cri/gn/<agenda>_loops.yml
loops:
  - id: L-014
    class: secrecy_vs_verification
    detected: 2025-07-02
    owner: <role>
    options: [RELAX, DETOUR, REBASE]
    decision: DETOUR
    rationale: "Confidential CO escrow w/ independent auditor"
    sla_deadline: 2025-10-01
    status: closed
    artifacts: ["/cri/co/Deterrence.yml#DET-CO5","/cri/bhr/registry.csv#Ω-002"]
```

**Pseudocode (skeleton).**

```
if loop_detected(L):
  pick = decide([RELAX, DETOUR, REBASE], evidence)
  log(L, pick, rationale)
  if deadline_passed and not closed:
     escalate(); consider REBASE default
```

**Anti-gaming.**

- *Looping to stall*: timebox + escalation.
- *Secrecy smokescreen*: require **perimeter COs** and exit tests.
- *Performative rebase*: demand side-by-side overlap window and continuity checks.

**Mini-examples.**

- *RELAX*: Discount band 1.5–3.5% + scenario dominance.
- *DETOUR*: Mutual confidential COs under escrow for hot-line latency.
- *REBASE*: Cross-border resolution adopts effects-based caps instead of platform caps.

---

# CARD 3 — Black-Hole Registry (BHR)

**Purpose.**
Declare zones **currently non-discussable** (security, privacy, sacred cores) **without disabling accountability**: publish risks, mitigations, **perimeter COs**, and **exit tests**.

**Inclusion criteria.**

- Disclosure would cause disproportionate harm *now*.
- Low-cost perimeter observables exist.
- A plausible **exit** or **sunset** can be stated.

**Required fields (one line per zone).**

- **ID & Name**
- **Scope** (what is dark)
- **Rationale** (why dark now)
- **Risks exported** (to whom)
- **Mitigations** (how risks are bounded)
- **Perimeter COs** (what remains visible)
- **Exit test / sunset** (what observation/date retires it)
- **Review cadence & owner**
- **Last review** (timestamp/version)

**CSV stub.**

```
id,name,scope,rationale,risks,mitigations,perimeter_cos,exit_test,review_cadence,owner,last_review

Ω-007,Model Weights,foundation models,IP+security,"asymmetric theft risk","post-incident audits; eval holdouts","AI-CO3; AI-CO5","time-delayed disclosure 18m",quarterly,Digital Regulator,2025-07-15
```

**Practices.**

- **No stigma**: registration ≠ wrongdoing.
- **Sunset discipline**: every entry has exit test/date or a renewed justification.
- **Interfaces**: publish what *can* be discussed around the core (perimeter COs).
- **Linkages**: BHR entries should reference DC plans and GN decisions they affect.

**Audit checks.**

- Backlog of past-due reviews = 0.
- Exit tests triggered? Retire promptly and version the public note.
- Perimeter COs actually live (data present, SLAs met).

**Mini-examples.**

- *Geoengineering readiness* (dual-use labs): perimeter COs = incident taxonomy counts; exit = independent monitoring online.
- *Covert sensors* (deterrence): perimeter COs = near-miss counts, hot-line latency; exit = post-program disclosure window.

---

## Cross-cutting Roles & Checklists

**Roles.**

- **Measurement Referee** — certifies COs, gates, hits; runs backcast/split-sample checks.

- **Protocol Marshal** — enforces DC/GN/BHR choreography; triggers collapse votes only on gate events.
- **Ledger Custodian** — maintains files, versions, hashes; publishes updates.

**One-page readiness check (per agenda, per quarter).**

```
reflexivity_statement: [Y/N]
co_pack_published: [Y/N]    audit_passed: [Y/N]
scale_map(v,I,L,back_maps): [Y/N]   losses_disclosed: [Y/N]
dc_plan: [Y/N]   timebox_ok: [Y/N]   ledger_current: [Y/N]
loops_open: <n>  on_time: [Y/N]
bhr_entries: <n> overdue_reviews: <n>=0?
cri: 0.__  confidence: 0.__  meets_thresholds: [Y/N]
decision: enact | continue DC | escalate GN | renew BHR | seek waiver
```

**Failure guard drills (quarterly).**

- DC closure dry-run on one agenda (simulate gate fire + counterfactual trace).
- GN escalation tabletop (expired SLA forces branch).
- BHR exit-test rehearsal (retire one entry end-to-end).

---

These cards turn the *idea* of reflexivity into **repeatable moves**: keep options live until the world speaks (DC), route impossibilities without pretense (GN), and name what must stay dark *without* going blind (BHR).

---

# Appendix C — Minimal Mathematical Formalism

*(state space, observables, invariants)*

## C1. State, Rules, and Reflexivity

**State space.**
Let $\mathcal{S} = \mathcal{S}_m \times \mathcal{S}_s \times \mathcal{S}_M$ denote micro/meso/macro components; $s = (x_m, x_s, x_M)$.

**Policies and rules.**
Policies $p \in \mathcal{P}$ are interventions. Rules/axioms $r \in \mathcal{R}$ specify evaluation, constraints, measurement, and disclosure.

**Reflexive agenda (formal).**
A policy $p$ is **self-referential** w.r.t. rules $r$ if applying $p$ updates $r$:

$$r^+ = T_r(p, r), \quad p \text{ is reflexive if } T_r(p, r) \neq r\,.$$

*Examples:* discount-rule changes, verification-protocol changes, jurisdiction remaps.

**Coupled dynamics.**

$$s_{t+1} = F(s_t, p_t, \varepsilon_t;\ r_t),$$
$$r_{t+1} = T_r(p_t, r_t),$$

with exogenous noise $\varepsilon_t$. Reflexivity means $F$ and $H$ below depend on $r_t$ that depends on the very $p_t$ under debate.

## C2. Observables and Collapse Observables (COs)

**Observation map.**
$H(\cdot\,; r)\colon \mathcal{S} \to \mathcal{Y}$, $y_t = H(s_t; r_t)$.

**Collapse Observable (CO).**
A CO is a pair $(h, g)$ with:

- $h: \mathcal{S} \to \mathbb{R}^d$ (a measurable feature),
- a **gate** $g: \mathbb{R}^{dW} \to \{0, 1\}$ acting on a window of length $W$.

**Fire condition.**
Given a history $\mathbf{z}_t = (h(s_t), \ldots, h(s_{t-W+1}))$, the CO **fires** when $g(\mathbf{z}_t) = 1$. A **CO kit** $K = \{(h_i, g_i)\}_{i=1}^m$ is **frame-bridging** if, for at least two competing frames $r^{(a)}, r^{(b)}$, the event $g_i = 1$ reduces the admissible policy set under both.

**Construct validity (minimal).** Each $h$ must specify: unit, window, method, audit path; each $g$ defines thresholds and persistence.

# C3. Decision Protocols as Operators

## (i) Delayed-Collapse (DC)

**Inputs.** Candidate set $C \subset \mathcal{P}$, alignment signals $A \subset K$, confidence $c_t \in [0, 1]$.
**Stopping time.**

$$\tau = \inf\{t: \exists (h_i, g_i) \in A \text{ s.t. } g_i(\mathbf{z}_t^{(i)}) = 1 \ \wedge \ c_t \geq \tau_c\}.$$

**Decision rule.** At $t = \tau$, pick $p_\star \in C$ specified by the gate $g_i$; publish counterfactuals for $C \setminus \{p_\star\}$.

## (ii) Gödel-Navigator (GN)

Let $\mathcal{G}$ be a directed graph over objects $\{p, r, y\}$ capturing "needed-for-evaluation" edges.
A **loop** exists if $\mathcal{G}$ has a directed cycle that prevents evaluation under current $r$. GN acts by:

- **RELAX**: $r \leftarrow R_\lambda(r)$ (weaken/band a rule; $\lambda$ controls slack).
- **DETOUR**: extend state/observation $(\mathcal{S}, \mathcal{Y}) \leftarrow (\mathcal{S} \times \widetilde{\mathcal{S}}, \mathcal{Y} \times \widetilde{\mathcal{Y}})$ via pilots/escrow to gather local evidence.
- **REBASE**: change the evaluation space (e.g., metric/jurisdiction) via $B$: $(\mathcal{S}, \mathcal{R}, \mathcal{Y}) \mapsto (\mathcal{S}', \mathcal{R}', \mathcal{Y}')$.

## (iii) Black-Hole Registry (BHR)

Partition $\mathcal{S}$ into visible $V$ and protected $\Omega$ (non-discussable now). Define **perimeter observables** $P: V \to \mathcal{Y}_\partial$ and an **exit stopping time** $\tau_{\text{exit}} = \inf\{t: \xi(y_{0:t}) \geq \theta\}$ that moves $\Omega$ content into $V$ or retires it.

# C4. Scale Maps, Invariants, and Losses

**Scale maps.**

$$M_{m \to s}: \mathcal{S}_m \to \mathcal{S}_s, \quad M_{s \to M}: \mathcal{S}_s \to \mathcal{S}_M,$$

with back-maps $M_{M \to s}^{-1}$, $M_{s \to m}^{-1}$ for implementation.

**Invariants $\mathcal{I}$.**
A set of functionals $\phi_j: \mathcal{S} \to \mathbb{R}$ such that for any forward/back-map $M$,

$$\phi_j(M(x)) = \phi_j(x) \quad \text{(e.g., conservation/budget/rights/time alignment).}$$

**Losses $\mathcal{L}$.**
Declare semantic losses (heterogeneity, timing, topology, privacy, phase). Optionally quantify information loss by mutual information drop:

$$L(M) = I(X;X) - I(M(X);X) = H(X) - I(M(X);X).$$

Loss statements must pair with mitigations (perimeter COs, tail percentiles, stress COs).

**CO hooks across scales.**
Each $(h_i, g_i)$ specifies hooks at micro/meso/macro: $h_i = (h_i^m, h_i^s, h_i^M)$ with declared invariants and losses per hook.

# C5. Stability and Early Warning (Curvature Proxies)

Model a local potential $V(x;\lambda)$ around a policy state $x$ (for intuition). Curvature given by Hessian $H = \nabla_x^2 V$. Approaching a tipping point effectively flattens curvature:

- variance ↑ (noise spreads further),
- AR(1) ↑ (slower decay),
- recovery rate $R \propto 1/\|\text{eig}(H)\| \downarrow$,
- flicker rate ↑ (boundary micro-crossings).

**EWI triplet.** Track {Var, AR1, Recovery} on core series; act only with **persistence** (over a window $W$) and **triangulation** (≥2 sources), and always via pre-declared CO gates.

# C6. Scoring and Confidence (Minimal Math)

**CRI.** For component vector $x = (sra, csc, co, dcr, gdn, bhc) \in [0,1]^6$ and weights $w \in \Delta^5$:

$$\text{CRI}(x;w) = \sum_{k=1}^{6} w_k x_k \,.$$

**Confidence.** $c = \frac{1}{2}\big(\text{completeness} + \text{reliability}\big)$, where reliability is a rescaled inter-rater statistic.

**Decision rule (default).** Enact if $\text{CRI} \geq \theta$ and $c \geq \tau_c$, else follow DC or GN path; exceptions logged with sunset.

# C7. Minimal Auditable Objects (schemas)

### CO row.

```
id: <CO-ID>
h:  "<state -> R^d definition>"
gate: "boolean on window W"
window: W
method: <audit path>
anti_gaming: [backcast, split_sample, honeypot]
hooks: {micro: <...>, meso: <...>, macro: <...>}
```

### Scale map.

```
invariants: [continuity,budget,rights,time]

losses: {heterogeneity: <mitigation>, topology: <mitigation>, privacy: <perimeter COs>, phase: <DC=true|
false>}

maps: {m_to_s: <method>, s_to_M: <method>, M_to_s: <obligation>, s_to_m: <obligation>}
```

### Loop docket (GN).

```
id: <L-id>  class: <type>  decision: RELAX|DETOUR|REBASE
rationale: <short>  sla: <deadline>  status: <open|closed>
```

**BHR entry.**

```
id: <Ω-id> scope: <hidden core> perimeter_cos: [<CO-ids>]
exit_test: <stopping rule> review: <cadence> owner: <role>
```

---

### What this formalism buys you

- A **stateful** picture where rules and observations can change with policy (reflexivity explicit).
- **Discussability** defined as existence of frame-bridging CO kits with gates.
- **Transportability** across scales via maps that preserve **invariants** and disclose **losses**.
- **Safe decision** via stopping times (DC), **loop routing** via GN operators, and **bounded opacity** via BHR.

It's minimal on purpose: small enough to run on a whiteboard during a council meeting—and precise enough that a stranger can audit your choices a century later.

---

# Appendix D — Audit Pack (checklists, logs, anti-gaming tests)

# D1. Master Audit Checklists

## D1.1 Pre-Launch (per agenda)

```
agenda: <name>    period: <YYYY-Q#>
reflexivity_statement: [Y/N]  link: <url>
co_pack:
  exists: [Y/N]  count: 5..9? [Y/N]
  preregistered: [Y/N]  anti_gaming: [Y/N]  hooks_across_scales: [Y/N]
scale_map:
  invariants_declared: [Y/N]  losses_declared: [Y/N]
  back_maps_owners_slas: [Y/N]
protocols:
  dc_plan_present: [Y/N]  timebox_defined: [Y/N]
  gn_docket_ready: [Y/N]  classes_defined: [Y/N]
  bhr_entries_listed: [Y/N]  exit_tests_set: [Y/N]
data:
  lineage_doc: [Y/N]  source_hashes: [Y/N]
  latency_sla_days: <n>  privacy_perimeter_cos: [Y/N]
signoff:
  measurement_referee: <name>  protocol_marshal: <name>  date: <date>
```

## D1.2 Quarterly CRI Integrity

```
portfolio: <name> period: <YYYY-Q#>
coverage:
  agendas_in_scope: <n>  agendas_scored: <n>  completeness: 0.__
reliability:
  method: kappa|ICC  value: 0.__  raters: <n>  blind_rescore_done: [Y/N]
weights:
  version: v__  changed_this_period: [Y/N]  backcast_invariance_pass: [Y/N]
components:
  floors_met: {co: [Y/N], gdn: [Y/N], bhc: [Y/N]}
decision_rule:
  CRI: 0.__  confidence: 0.__  meets_thresholds: [Y/N]
exceptions:
  waivers_active: <n>  all_timeboxed: [Y/N]  counterfactual_traces_published: [Y/N]
audit_result: PASS | PASS_w_mitigations | FAIL
```

## D1.3 CO Kit Operation

```
co_pack_id: <CO-PACK-ID>

design_checks: [prereg, auditability, frame_bridging, anti_gaming, cross_scale_hooks]=[Y/N x5]
```

```
run_checks:
  gate_definitions_precise: [Y/N]
  gate_hits_logged: [Y/N]  persistence_window_met: [Y/N]
  triangulation_sources>=2: [Y/N]  latency_sla_met: [Y/N]
anti_gaming_suite_run: [Y/N]  failures: <list or "none">
status: PASS | PASS_w_mitigations | FAIL
```

### D1.4 DC (Delayed-Collapse) Gate Verification

```
dc_plan: <url>  timebox: <T>  candidates: [Pi]
alignment_signals: [Aj<-CO]  gates: [Gk]
ledger_current: [Y/N]  gate_fired?: [Y/N]
confidence_c>=tau?: [Y/N]
counterfactual_trace_published: [Y/N]
if timebox_expired_without_gate: escalated_to_gn?: [Y/N]
status: PASS | PASS_w_mitigations | FAIL
```

### D1.5 GN (Loop Docket) Compliance

```
open_loops: <n>  on_time_fraction: 0.__
each(loop):
  class: <type>  owner: <role>  decision: RELAX|DETOUR|REBASE
  deadline_met: [Y/N]  rationale_logged: [Y/N]  artifacts_linked: [Y/N]
escalations_done: [Y/N]
status: PASS | PASS_w_mitigations | FAIL
```

### D1.6 BHR (Black-Hole Registry) Hygiene

```
entries: <n>  overdue_reviews: <n>=0?  perimeter_cos_live: [Y/N]
exit_tests_defined_all: [Y/N]  retired_entries_this_period: <n>
sunset_discipline: PASS|WARN|FAIL
status: PASS | PASS_w_mitigations | FAIL
```

### D1.7 Scale-Map Validation

```
map_version: v__ invariants_respected: [continuity,budget,rights,time,attribution]=[Y/N×5]

loss_mitigations_present: [heterogeneity,topology,privacy,phase]=[Y/N×4]

back_map_execution_spot_checks:
  samples: <n>  pass_rate: 0.__  issues: <list or "none">
status: PASS | PASS_w_mitigations | FAIL
```

### D1.8 EWI Discipline (Early-Warning)

```
series: <id>  baseline:{var:__,AR1:__,recovery:__}
current: {var:__,AR1:__,recovery:__}  flicker_count: <n>
triangulation>=2: [Y/N]  persistence>=W: [Y/N]
gate_tied_to_policy: [Y/N]  alarm_log_updated: [Y/N]
status: PASS | PASS_w_mitigations | FAIL
```

### D1.9 Data Pipeline & Privacy

```
sources: [..]  lineage_graph_url: <url>  source_hashes_verified: [Y/N]
latency_sla_days: <n>  breaches_this_period: <n>
privacy_controls:
  deidentification: [Y/N]  perimeter_cos_instead_of_raw: [Y/N]
penetration_test_last:<date>  issues_open:<n>
status: PASS | PASS_w_mitigations | FAIL
```

---

# D2. Standard Logs (machine-readable stubs)

### D2.1 Audit Log (master)

```
audit_id,period,scope,auditor,started,ended,result,severity,findings_link,fix_by,owner,status
```

A-2025Q3-CRI,2025Q3,CRI_Integrity,ExtPanel,2025-07-10,2025-07-25,PASS,,/packs/CRI_2025Q3.pdf,,SecRefGov,closed

A-2025Q3-CO-CLM,2025Q3,CO_Climate,IntTeam,2025-07-05,2025-07-12,PASS_w_mitigations,MED,/packs/CO_CLM.pdf,2025-08-15,ClimateUnit,open

## D2.2 Anti-Gaming Findings

id,agenda,test_id,date,hypothesis,observed,pass?,severity,action,owner,fix_due

AG-031,Finance,THRESH_BUNCH,2025-07-08,"values bunch at gate",KS p=0.01,false,HIGH,"raise audit; randomize audits","SecFin",2025-08-01

AG-044,AI,PROXY_SWAP,2025-07-20,"risk metric swapped",detected,true,LOW,"lock rubric; add cross-metric check","AIReg",2025-07-30

## D2.3 Gate Verification (DC)

gate_id,agenda,date,inputs_ok?,persist_ok?,triang_ok?,confidence,gate_fired?,collapsed_to,trace_link,status

G-CLM-02,Climate,2025-07-31,TRUE,TRUE,TRUE,0.78,TRUE,P4,/traces/CLM_P4_2025Q3.pdf,closed

## D2.4 Loop Docket (GN)

loop_id,agenda,class,detected,owner,decision,deadline,status,rationale,artifacts

L-014,Deterrence,secrecy_vs_verification,2025-07-02,MOD,DETOUR,2025-10-01,closed,"confidential CO escrow","/co/Det.yml#DET-CO5;/bhr/Ω-002"

## D2.5 BHR Registry

id,name,scope,risks,perimeter_cos,exit_test,review_cadence,owner,last_review,next_review

Ω-007,Model Weights,AI core,"IP/security","AI-CO3;AI-CO5","time-delayed disclosure 18m",quarterly,DigitalReg,2025-07-15,2025-10-15

## D2.6 Waiver Ledger

waiver_id,agenda,justification,sunset,audit_intensity,signoff,trace_link,status

W-2025-03,Deterrence,"imminent risk posture",2026-01-01,monthly,"Chair+OppLead","/waivers/W-2025-03_trace.pdf",active

## D2.7 Metric/Weight Change Log

change_id,object,type,old,new,date,rationale,backcast_delta,owner

MW-2025-12,weights,CRI,"eq","{sra:0.2,csc:0.2,co:0.2,dcr:0.15,gdn:0.15,bhc:0.1}",2025-12-15,"focus on measurement","mean +0.02","SecRefGov"

## D2.8 Backcast Test Log

run_id,object,periods,delta_mean,anomalies,result,report_link

BC-2025Q3-CRI,CRI,2019Q1..2025Q2,+0.03,"2021Q4 Finance spike",PASS,/reports/BC_CRI_2025Q3.pdf

## D2.9 Blind Re-score Log

id,agenda,component,raterA,raterB,delta,threshold,adjudication,decision,notes

BR-112,Finance,co,0.66,0.33,0.33,0.30,required,"0.50","anti-gaming incomplete; plan logged"

---

# D3. Anti-Gaming Test Suite (v1)

Run suite per agenda **each quarter**; record into **D2.2**.

| Test ID | Name | Hypothesis Detected | Procedure | Pass Criteria |
|---|---|---|---|---|
| BACKCAST | Backcast invariance | Thresholds tuned post hoc | Apply current thresholds to past windows; compare hit rates | Δ hit rate within pre-declared band (e.g., ≤ ±5%) unless justified |
| SPLIT_STAB | Split-sample stability | Selection bias | Train/threshold on split A; validate on B (region/time/provider) | Comparable metrics across splits (t-test/KS within band) |
| PLACEBO | Placebo/ honeypot | Paper compliance | Include a metric that should not move; flag if it does | Placebo stays flat (no systematic shift) |
| WIN_LOCK | Window pre-reg lock | Window shopping | Hash & timestamp window spec; verify at audit | Hash match; deviations logged & justified |
| LAT_SLA | Latency SLA | Latency doping | Compare actual lags vs SLA; alert on breaches | ≥95% within SLA; breaches investigated |
| SUB_WATCH | Substitution watch | Proxy channels proliferate | Monitor near-substitute volumes when a metric tightens | No unexplained surge; if surge, perimeter CO added |
| THRESH_BUNCH | Threshold bunching | Values bunch at gate | Bunching test (McCrary/KS) around cutoffs | No significant bunching; if yes, trigger random audits |
| XMET_TENSION | Cross-metric tension | Output “improves” while inputs don’t | Correlate necessary inputs vs claimed outcomes | Consistent movement; else investigate |
| SIMPSON | Simpson’s paradox scan | Aggregation hide subgroup harm | Compare aggregate trend vs subgroup trends | No paradox; or disclose & adjust decision scope |
| COUNTERFACT | Counterfactual consistency | Collapse chosen contradicts trace | Simulate alt candidates using observed A | Chosen collapse aligns with gate & trace |
| EVAL_ROT | Eval rotation (AI) | Overfitting to known evals | Rotate/hold-out evals; external auditors | No suspicious jumps only on public evals |
| TOP_STRESS | Topology stress (finance) | Masked network risk | Stress hubs vs average; compare spread | Hubs consistent; disclose if divergence |

**Implementation notes.**

- Pre-declare bands & alpha levels (e.g., $p<0.05$) in `/cri/policy/tests.yml`.
- Publish **failures and fixes** with deadlines.

---

# D4. Runbooks (who does what, when)

## D4.1 Quarterly Audit Calendar

```
week_1: ["CRI integrity audit","Backcast tests","Blind re-score"]
week_2: ["CO kit ops audits","EWI discipline checks"]
week_3: ["DC gate verification","GN docket SLA review"]
week_4: ["BHR hygiene","Data pipeline & privacy","Red-team debrief"]
publish: "dashboard refresh + audit summary PDF"
owners:
  measurement_referee: <name>
  protocol_marshal: <name>
  ledger_custodian: <name>
```

## D4.2 Severity & Remediation

```
severity_scale:
  HIGH: "Decision-critical failure or evidence of gaming"
  MED:  "Material weakness; mitigations available"
  LOW:  "Minor process gap"
remediation_sla_days:
  HIGH: 30
  MED:  60
  LOW:  90
```

---

# D5. One-Page Audit Summary (public)

```
period: 2025Q3
headline: "CRI meets enactment gates; two mitigations open"
scores: {CRI: 0.64, confidence: 0.74}
passes:
  - CRI integrity PASS
  - DC gate verification PASS (Climate → P4)
warns:
  - CO(CLIMATE): split-sample stability incomplete → due 2025-08-15
  - Finance: threshold bunching detected near LCR cut-off → random audits activated
exceptions:
  - Waiver W-2025-03 (Deterrence) active, sunset 2026-01-01, monthly audits
redteam:
  - Proxy swapping attempt detected (AI) → cross-metric check added
links:
  - /cri/score/CRI_timeline.csv
  - /cri/dc/Climate_ledger.csv
  - /cri/gn/loops.csv
  - /cri/bhr/registry.csv
```

---

# D6. Quick Door-Check (meeting use)

Before any vote or public communiqué:

- Gate **actually fired** (DC) **and** counterfactual trace attached
- No **open HIGH** severity audit items on this agenda
- **CRI & c** above thresholds; component floors met (co, gdn, bhc)
- If **waiver**, ledger + sunset + intensified audit agreed
- Updated artifacts pushed & hashed on dashboard

---

This Audit Pack turns slogans into **receipts**: who measured what, when, with which guardrails—and what you did when the numbers tried to lie back.

---

# Appendix E — Glossary & Canon (terms that should not drift)

## E1. Core Concepts

- **Axiomatic Reflexivity** (`axiomatic_reflexivity`, *type: concept*)
  Governance discipline that requires axioms (A1–A4), not ideologies, to make self-referential agendas discussable.
  *Do-not-drift:* Not a prediction theory; it binds procedures and observables.

- **Self-Referential Agenda** (`sra_agenda`, concept)
  A policy whose adoption changes the rules/metrics by which it is evaluated.
  *Do-not-drift:* Include scope (payoffs, constraints, information).

- **Self-Referential Attractor (SRA)** (`sra_attractor`, dynamic)

Feedback basin where beliefs/actions reinforce the rules that justify them.
*Do-not-drift:* Distinct from "agenda"; it's a system state, not a proposal.

- **Discussability** (`discussability`, criterion)
  The condition that at least one **CO** is bound and frame-bridging.
  *Do-not-drift:* No CO → rhetoric, not discussable governance.

- **Non-Phase-Free Regime** (`non_phase_free`, property)
  Outcomes depend on order/timing; early commitment risks path dependence.
  *Do-not-drift:* Triggers **Delayed-Collapse** by default.

# E2. Axioms (Postulates)

- **A1 Reflexivity** (`A1`) Some agendas change their evaluation rules.

- **A2 Observability** (`A2`) Claims are discussable only if they bind ≥1 **CO**.

- **A3 Incompleteness** (`A3`) Some disputes are undecidable without axiom expansion or domain rebasing.

- **A4 Delayed-Collapse** (`A4`) Keep ≥2 candidates live in non-phase-free regimes until alignment gates fire.
  *Do-not-drift:* Axioms are procedural commitments, not policy preferences.

# E3. Measurement Objects

- **Collapse Observable (CO)** (`co`, object)
  Pair `(h,g)` where `h` measures; `g` fires on a window; pre-registered, auditable, frame-bridging.
  *Do-not-drift:* COs are not "KPIs" unless they meet pre-reg + gate + audit.

- **CO Kit** (`co_kit`, bundle)
  Set of 5–9 COs with methods, gates, anti-gaming, cross-scale hooks.
  *Do-not-drift:* Fewer than 5 rarely frame-bridge; more than 9 invites gaming.

- **Gate (fires)** (`gate`, event)
  Boolean condition on CO history that, if met with confidence $c \geq \tau$, authorizes collapse/action.

- **Alignment Signal** (`align_signal`, signal)
  CO-tied metric used to decide between candidates under **Delayed-Collapse**.

- **Scale-Map** (`scale_map`, mapping)
  Forward/back mappings across micro↔meso↔macro with **invariants** and **losses** declared, plus back-map obligations (owners, SLAs).

- **Invariant** (`invariant`, property)
  Quantity/constraint preserved by a scale-map (continuity, budget, rights, time alignment, attribution).

- **Loss** (`loss_declared`, property)
  What mapping discards (heterogeneity, timing, topology, privacy, phase) + mitigations (perimeter COs, tail COs).

- **Perimeter CO** (`perimeter_co`, object)
  Observable around a protected core that keeps accountability live without exposing the core.

# E4. Index & Confidence

- **Civic Reflexivity Index (CRI)** (`cri`, index)
  Weighted sum over six components in `[0,1]`: `sra, csc, co, dcr, gdn, bhc`.
  *Do-not-drift:* CRI ≠ approval; it's capability to govern reflexively.

- **CRI Components**

- `sra`: Self-Referential Attractor density (explicit reflexivity statements with scope).
- `csc`: Cross-Scale Consistency (map quality with invariants/losses/back-maps).
- `co`: CO Pack Rate (share of agendas with valid kits).
- `dcr`: Delayed-Collapse Rate (share of non-phase-free agendas under DC).
- `gdn`: Gödel-Navigator usage (loop detection & routing quality).
- `bhc`: Black-Hole Coverage (registry completeness with exit tests).

- **Confidence (`c`)** (`confidence_c`, index)
  Mean of completeness and inter-rater reliability, both in `[0,1]`.
  *Do-not-drift:* Always publish with CRI; low `c` voids enactment.

# E5. Protocols

- **Delayed-Collapse (DC)** (`protocol_dc`, protocol)
  Keep candidates `C`; observe alignment signals; collapse on pre-declared gates with $c \geq \tau$; publish counterfactual trace.
  *Do-not-drift:* DC is not "indecision"; it's structured waiting with a timebox.

- **Gödel-Navigator (GN)** (`protocol_gn`, protocol)
  Handles undecidable loops via **RELAX** (weaken/band axiom), **DETOUR** (pilot/escrow), **REBASE** (change metric/venue).
  *Do-not-drift:* Must log class, owner, SLA, decision, rationale.

- **Black-Hole Registry (BHR)** (`protocol_bhr`, registry)
  Declares currently non-discussable zones with risks, mitigations, perimeter COs, exit tests, review cadence.
  *Do-not-drift:* Registration ≠ guilt; every entry needs sunset or exit test.

# E6. Audit & Integrity

- **Backcast Invariance** (`test_backcast`, test)
  Today's thresholds applied to past data yield stable hit rates unless justified.

- **Split-Sample Stability** (`test_split_stability`, test)
  Metrics behave similarly across held-out partitions.

- **Honeypot / Placebo Metric** (`test_placebo`, test)
  A metric that should not change; movement flags gaming.

- **Substitution Watch** (`test_substitution`, test)
  Monitor near-substitutes when a regulated metric tightens.

- **Threshold Bunching** (`test_bunching`, test)
  Detects value spikes around a cutoff (manipulation).

- **Window Pre-Registration** (`test_winlock`, control)
  Hash/timestamp analysis windows to prevent after-the-fact shifts.

- **Latency SLA** (`latency_sla`, control)
  Published data lag limits; breaches trigger audits.

- **Counterfactual Trace** (`counterfactual_trace`, artifact)
  Evidence bundle showing how non-chosen candidates would have performed under observed signals.

- **Waiver** (`waiver_ledger`, exception)
  Time-boxed exception to enactment thresholds with intensified audit and counterfactual trace plan.

# E7. Early-Warning & Stability

- **Variance (EWI-Var)** (`ewi_var`, indicator)
  Windowed variance increase as curvature flattens.

- **Recovery (EWI-Rec)** (`ewi_recovery`, indicator)
  Time to return to baseline after small shocks (slower is worse).

- **Lag-1 Autocorrelation (EWI-AR1)** (`ewi_ar1`, indicator)
  Closer to 1 indicates slowing recovery.

- **Flicker** (`ewi_flicker`, indicator)
  Validated micro-switches across a declared boundary.
  *Do-not-drift:* EWIs warn; they don't decide—tie to CO gates with persistence + triangulation.

# E8. Roles & Process

- **Measurement Referee** (`role_referee`, role)
  Certifies COs, gates, hits; runs backcast/split-sample checks.

- **Protocol Marshal** (`role_marshal`, role)
  Enforces DC/GN/BHR choreography; triggers collapse votes only on gate events.

- **Ledger Custodian** (`role_custodian`, role)
  Maintains artifacts, versions/hashes, publishes dashboard.

- **Collapse Vote** (`collapse_vote`, event)
  A decision event permitted only upon verified gate fire and counterfactual trace presentation.

# E9. Ethics & Disclosure

- **Perimeterization** (`eth_perimeter`, practice)
  Keep cores dark while publishing perimeter COs for accountability.

- **Reciprocity** (`eth_reciprocity`, principle)
  Balance verification exposure via symmetrical or escrowed COs.

- **Sunset/Exit Test** (`eth_sunset`, control)
  Every secrecy claim expires or proves continuance via observation.

- **Epistemic Humility** (`eth_humility`, norm)
  Publish uncertainty/confidence with equal prominence.

---

# E10. Canon (machine-readable stub)

```
# /cri/canon/v1.0.yml
version: "1.0"
since: "2025-08-29"
terms:
  - key: axiomatic_reflexivity
    name: "Axiomatic Reflexivity"
    type: concept

def: "Procedure-first discipline (A1–A4) that makes self-referential agendas discussable by binding
observables and protocols."

    notes: "Not a theory of outcomes; a measurement contract."
    since: "2025-08-29"
  - key: co
    name: "Collapse Observable"
    type: object

def: "Pre-registered, auditable (h,g) pair; gate fires on window to narrow policy space across frames."

    notes: "Must be frame-bridging; includes anti-gaming."
    since: "2025-08-29"
```

```
  - key: cri
    name: "Civic Reflexivity Index"
    type: index
    def: "Weighted sum over {sra,csc,co,dcr,gdn,bhc} in [0,1]; capability to govern reflexively."
    notes: "Always publish confidence c; floors apply."
    since: "2025-08-29"
  - key: protocol_dc
    name: "Delayed-Collapse"
    type: protocol

def: "Keep ≥2 candidates until alignment gates fire with confidence; then collapse and publish
counterfactual trace."

    notes: "Timeboxed; reopen only via GN."
    since: "2025-08-29"
  - key: protocol_gn
    name: "Gödel-Navigator"
    type: protocol

def: "Routes undecidable loops via RELAX/DETOUR/REBASE with SLA and rationale."

    notes: "Loop docket required."
    since: "2025-08-29"
  - key: protocol_bhr
    name: "Black-Hole Registry"
    type: registry

def: "Declares non-discussables with risks, mitigations, perimeter COs, exit tests, and review cadence."

    notes: "Registration has no stigma; sunset discipline."
    since: "2025-08-29"
  - key: ewi_var
    name: "EWI–Variance"
    type: indicator

def: "Windowed variance used as early-warning under persistence+triangulation."

    notes: "Never sole trigger; tie to CO gate."
    since: "2025-08-29"
  # … include remaining keys from sections E1–E9 …
```

---

## Editor’s note (anti-drift checklist)

- New term maps to existing canon? If not, add with version bump.
- Abbrev unique? Checked against reserved list.
- Backcast note prepared if change affects scoring/decisions.
- Cross-refs updated (CO kits, DC/GN/BHR cards, audit pack).

This glossary & canon is the **semantic operating system** of the handbook. Keep it tight, version it like code, and the rest of the architecture will stay discussable.

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This work is speculative, interdisciplinary, and exploratory in nature. It bridges metaphysics, physics, and organizational theory to propose a novel conceptual framework—not a definitive scientific theory. As such, it invites dialogue, challenge, and refinement.

*I am merely a midwife of knowledge.*